Chapter 0-2
JDK & Eclipse Setup

What Is the Java™ Technology?

- Java technology is:
  - A programming language
  - A development environment
  - An application environment
  - A deployment environment
  - It is similar in syntax to C++.
  - It is used for developing both *applets* and *applications*.

History

- Interactive Television Project
  - James Gosling, Mike Sheridan, and Patrick Naughton initiated the Java language project in June 1991
  - ***Oak*** after an oak tree that stood outside Gosling's office
  - Later the project went by the name ***Green*** and was finally renamed ***Java***

History



- Java
  - Sun Microsystems released the first public implementation as Java 1.0 in 1995
    - It promised "Write Once, Run Anywhere"
  - released initially as J2SE 1.2 in December 1998 – 1999
    - new versions had multiple configurations built for different types of platforms

History

- Different types of platforms
  - JAVA EE
    - *J2EE* included technologies and APIs for enterprise applications typically run in server environments
  - *Java ME*
    - *J2ME* featured APIs optimized for mobile applications
  - Java SE
    - The desktop version was renamed *J2SE*

Principles

- There were five primary goals in the creation of the Java language
  - It must be "simple, object-oriented and familiar"
  - It must be "robust and secure"
  - It must be "architecture-neutral and portable"
  - It must execute with "high performance"
  - It must be "interpreted, threaded, and dynamic"

Versions

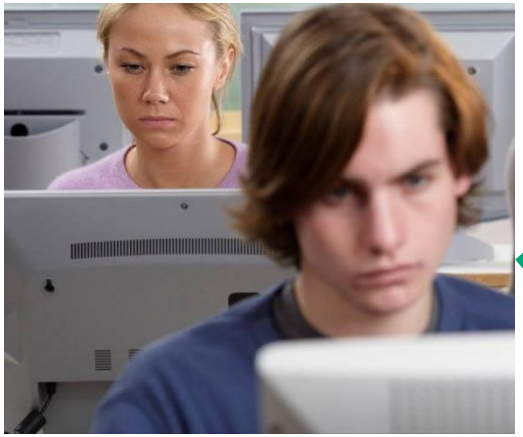
- Major release versions of Java
  - JDK 1.0 (January 21, 1996)
  - JDK 1.1 (February 19, 1997)
  - J2SE 1.2 (December 8, 1998)
  - J2SE 1.3 (May 8, 2000)
  - J2SE 1.4 (February 6, 2002)
  - J2SE 5.0 (September 30, 2004)
  - Java SE 6 (December 11, 2006)
  - Java SE 7 (July 28, 2011)
  - Java SE 8 (March 18, 2014)

Java platform

- **Compiled Language**
  - C
  - 通过编译器一次将源代码翻译成机器可以识别的目标代码，今后运行的都为目标代码

Java platform

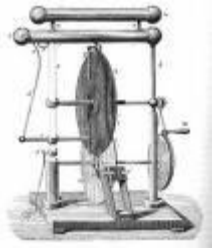
- **Interpreted language**
  - Basic、Perl
  - 代码每次运行时由解释器将其翻译成目标代码，一边解释一边运行



You

write

Perl Program



Perl interpreter runs it – things happen

Java platform

- Cross-platform computing environment
  - programs written for the Java platform must run similarly on any combination of hardware and operating system with adequate runtime support
  - **Semi-Compiled**
    - converts ***Java source code*** into ***Java bytecode***
  - **Semi-Interpreted**
    - ***a "virtual machine"*** that executes ***Java bytecode programs***

Java platform

- Java bytecode
  - analogous to machine code
  - be executed by a virtual machine
- Machine code
  - architecture-specific

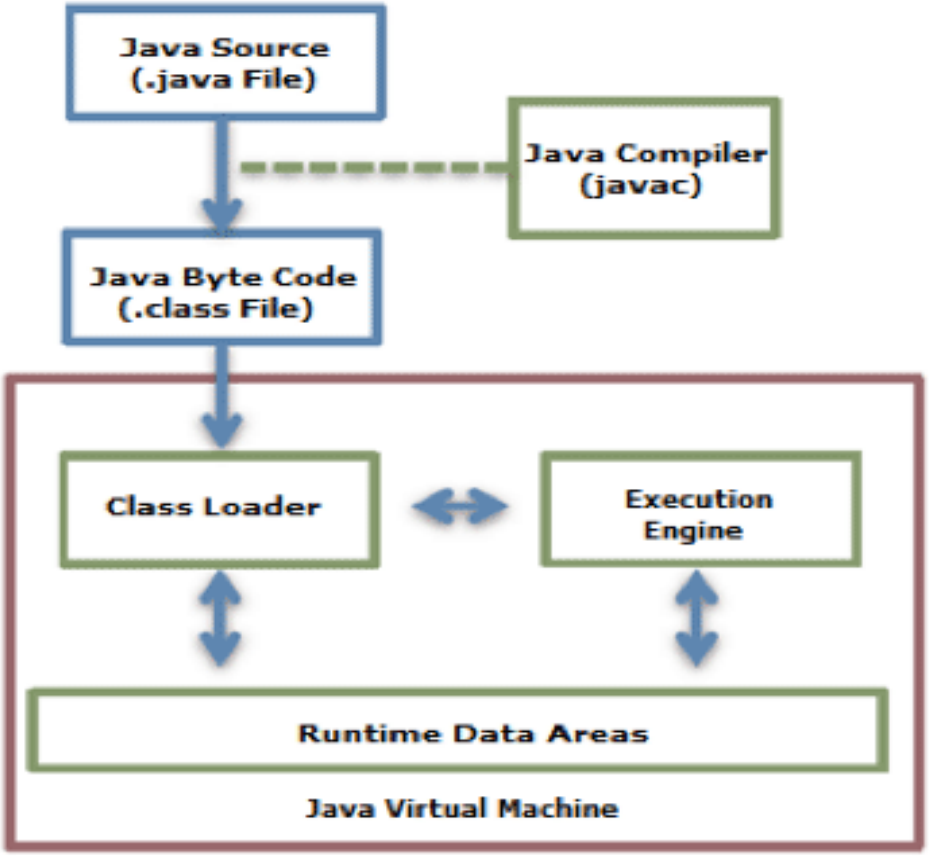
Java platform

- **Java Virtual Machine(JVM)**
  - The heart of the Java platform
  - The JVM has the following major subsystems
    - **Class loader**
    - **Execution Engine**

Java platform

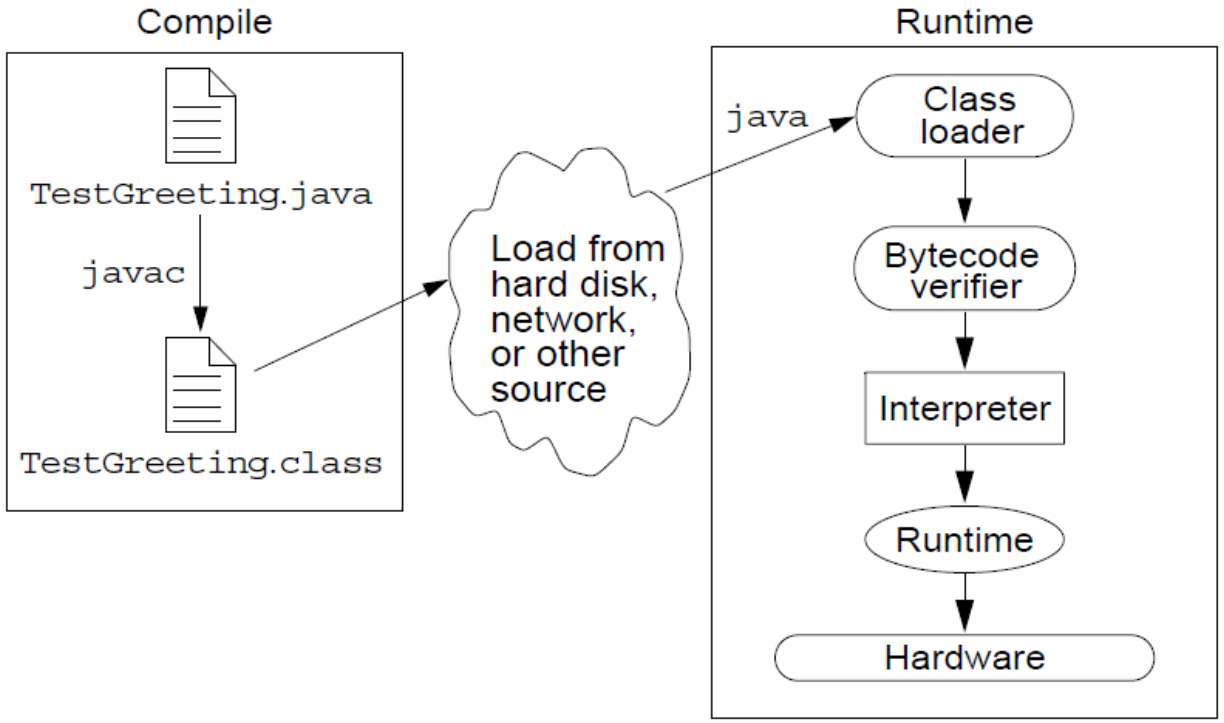
- **Class loader**
  - Responsible for loading all classes necessary for the execution of a program into the data areas.
- **Execution Engine**
  - Responsible for executing instructions from the data areas.
    - **Bytecode interpreter**
    - **(JIT)just-in-time compiler**

Java platform



Java platform

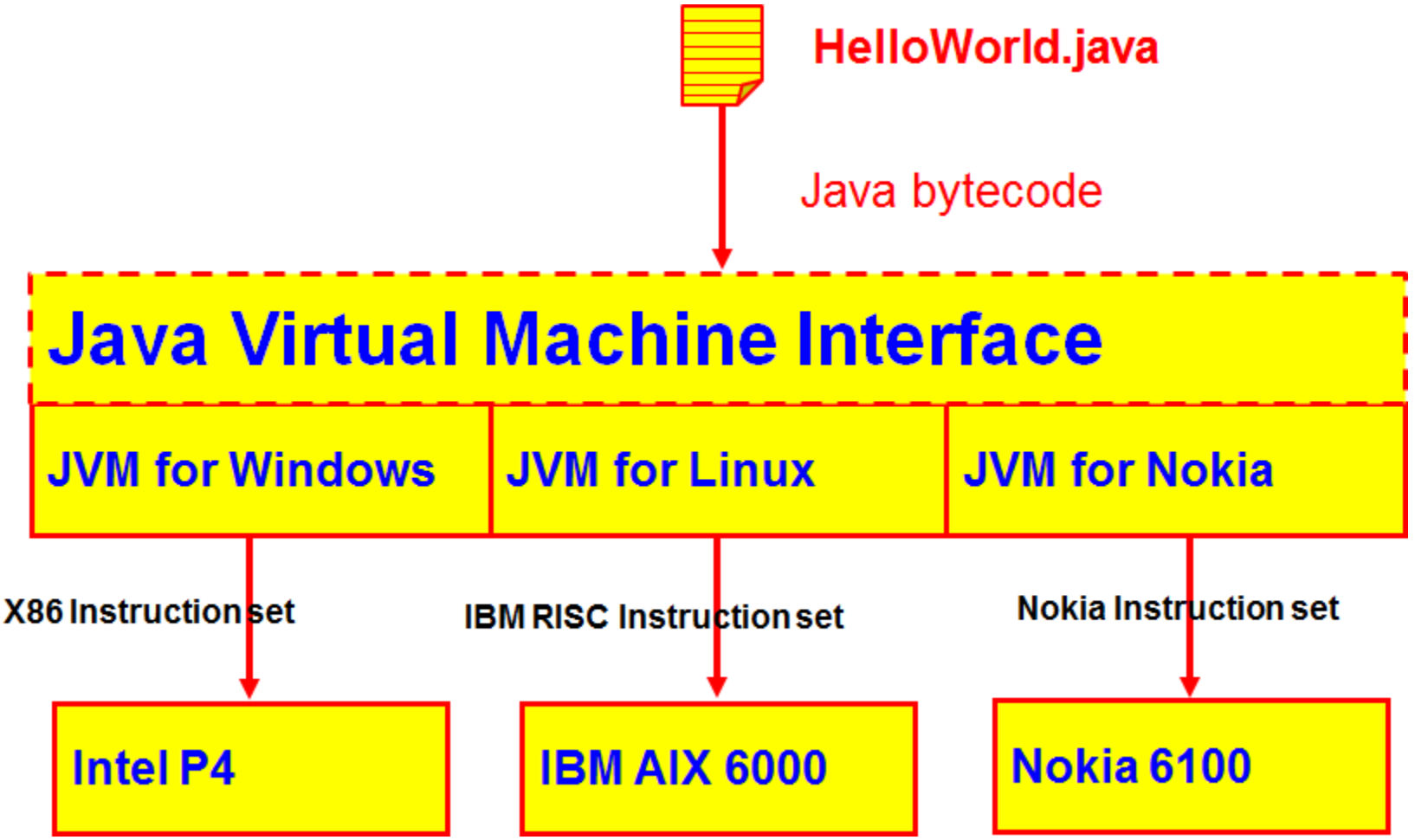
- The Java application environment performs as follows:



Java platform

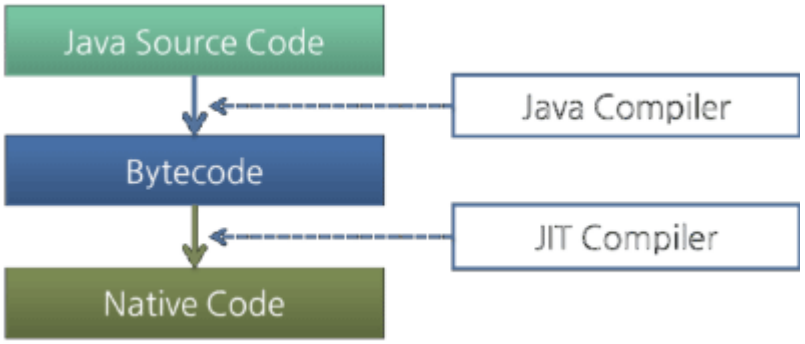
- Programs written in Java have a reputation for being slower and requiring more memory than those written in C++.
- Java was designed to run based on a virtual machine separated from a physical machine for implementing **WORA** (*Write Once Run Anywhere*)

Java platform

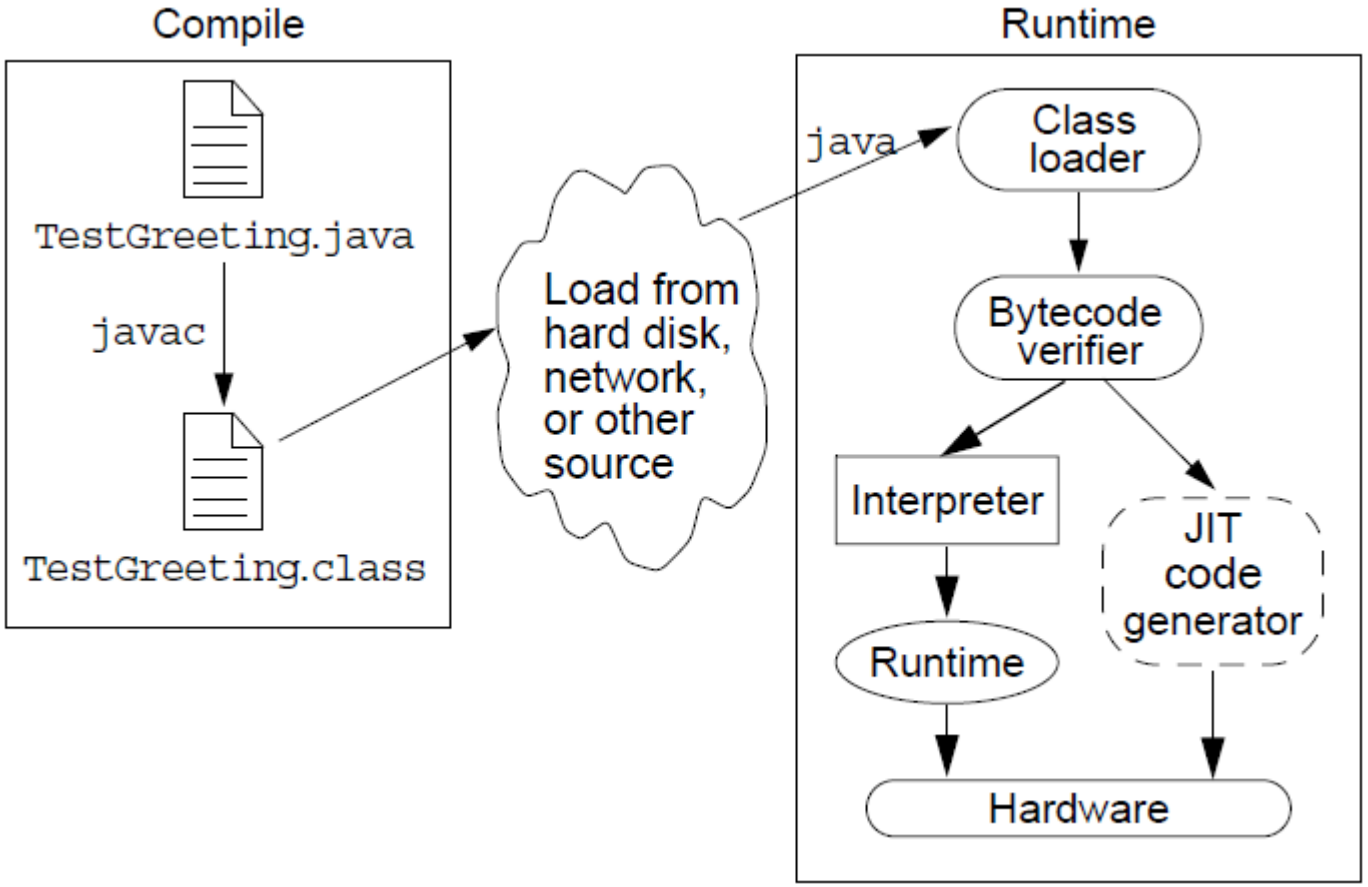


Java platform

- JIT(Just In time) Complier
  - translates the Java bytecode into native processor instructions at run-time and caches the native code in memory during execution



Java platform



JDK

- **Java Development Kit (JDK)**
  - An implementation of either one of the Java SE, Java EE or Java ME platforms
    - a complete Java Runtime Environment(JRE)
    - a collection of programming tools
    - a wide selection of example programs demonstrating the use of almost all portions of the Java API

JRE

- JRE(Java Runtime Environment)
  - consists of a **Java Virtual Machine**
  - all of the **class libraries** present in the production environment, as well as **additional libraries** only useful to developers

JDK Download

- JDK Download

http://www.oracle.com/technetwork/java/javase/downloads/index-jsp-138363.html

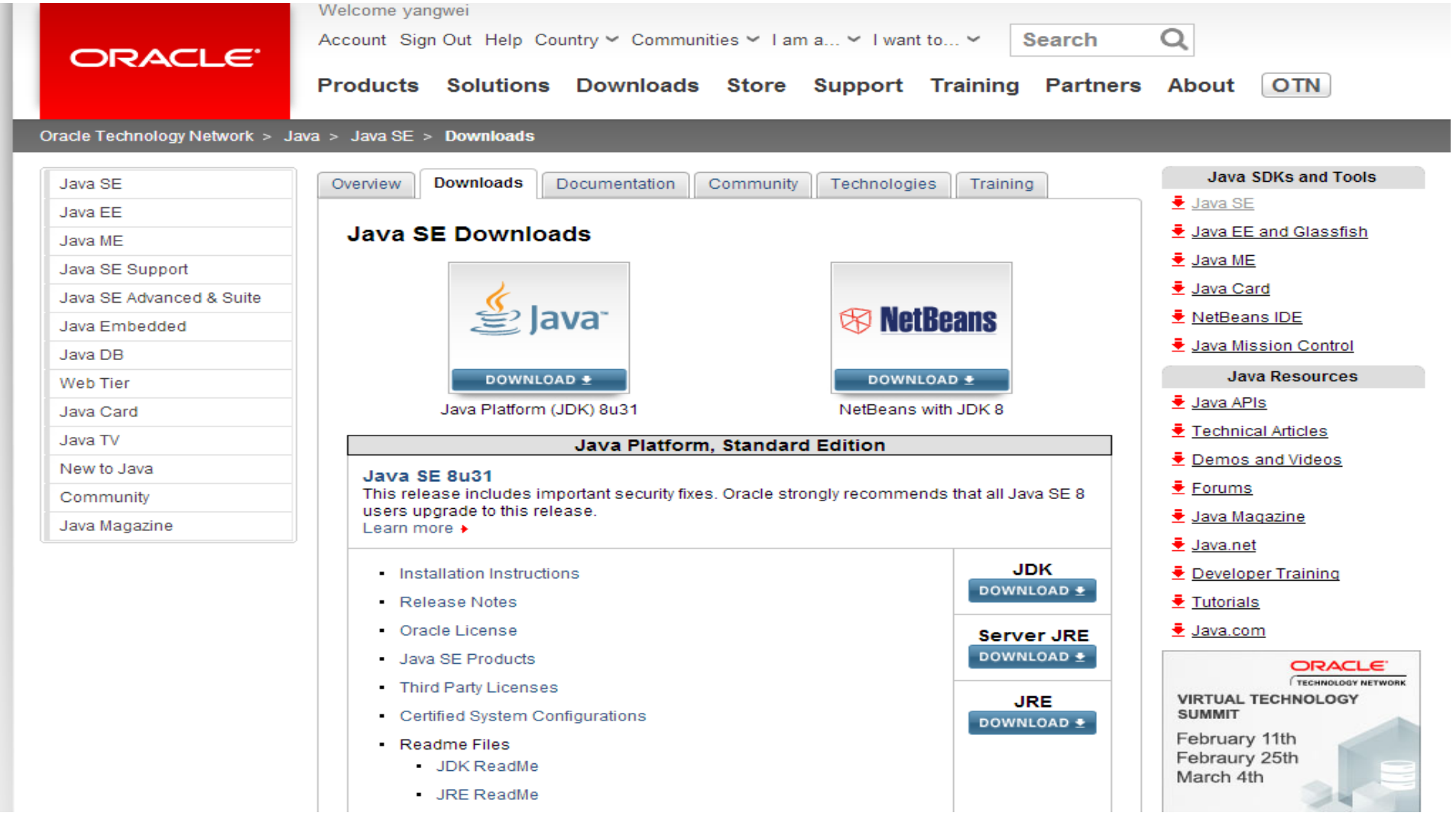
- JDK install

jdk-8u31-windows-x64.exe(window 64 bit)

jdk-8u31-windows-i586.exe(window 32 bit)

- JDK deployment

JDK Download



- Java ME
- Java SE Support
- Java SE Advanced & Suite
- Java Embedded
- Java DB
- Web Tier
- Java Card
- Java TV
- New to Java
- Community
- Java Magazine

Java SE Development Kit 8 Downloads

Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications, applets, and components using the Java programming language.

The JDK includes tools useful for developing and testing programs written in the Java programming language and running on the Java platform.

See also:

- Java Developer Newsletter (tick the checkbox under Subscription Center > Oracle Technology News)
- Java Developer Day hands-on workshops (free) and other events
- Java Magazine

JDK MD5 Checksum

Looking for JDK 8 on ARM?
JDK 8 for ARM downloads have moved to the JDK 8 for ARM download page.

Java SE Development Kit 8u31

You must accept the Oracle Binary Code License Agreement for Java SE to download this software.

○ **Accept License Agreement** ◉ **Decline License Agreement**

Product / File Description	File Size	Download
Linux x86	135.24 MB	jdk-8u31-linux-i586.rpm
Linux x86	154.91 MB	jdk-8u31-linux-i586.tar.gz
Linux x64	135.62 MB	jdk-8u31-linux-x64.rpm
Linux x64	153.45 MB	jdk-8u31-linux-x64.tar.gz
Mac OS X x64	209.17 MB	jdk-8u31-macosx-x64.dmg
Solaris SPARC 64-bit (SVR4 package)	136.91 MB	jdk-8u31-solaris-sparcv9.tar.Z
Solaris SPARC 64-bit	97.11 MB	jdk-8u31-solaris-sparcv9.tar.gz
Solaris x64 (SVR4 package)	137.51 MB	jdk-8u31-solaris-x64.tar.Z
Solaris x64		
Windows x86	157.96 MB	jdk-8u31-windows-i586.exe
Windows x64	170.36 MB	jdk-8u31-windows-x64.exe

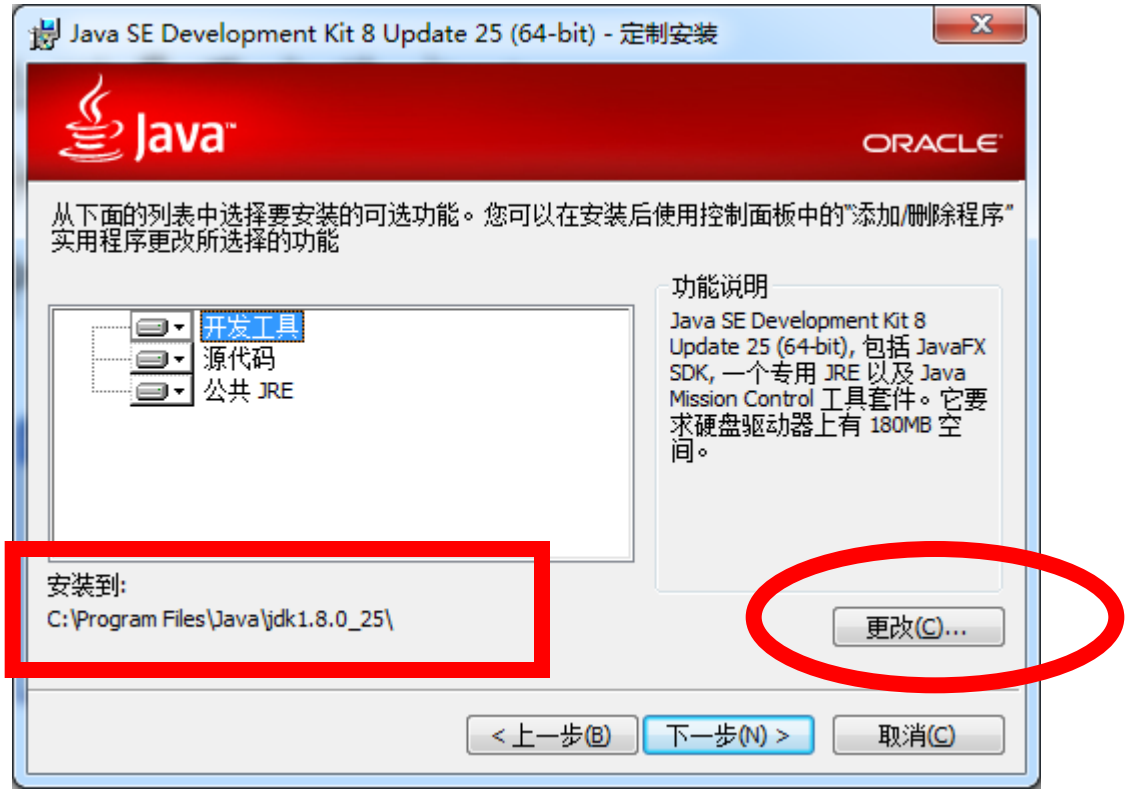
JDK Install

- 运行JDK安装程序，进入安装过程



JDK Install

- 选择安装目录



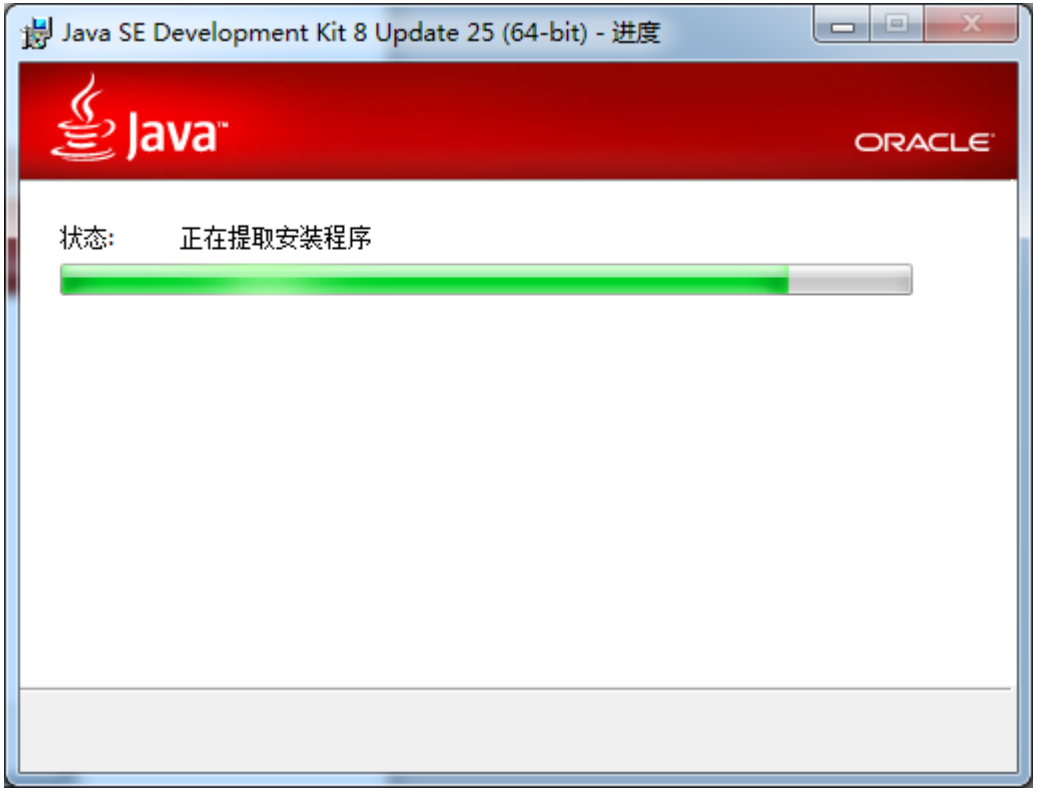
JDK Install

- 更改安装目录



JDK Install

- 安装进行中



JDK Install

- Install JRE



JDK Install

- 选择安装目录



JDK Install

- 安装进行中



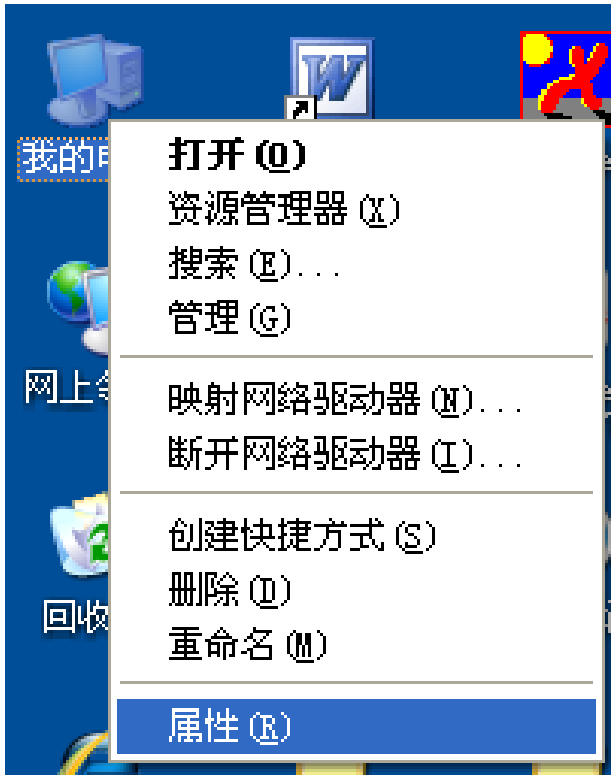
JDK Install

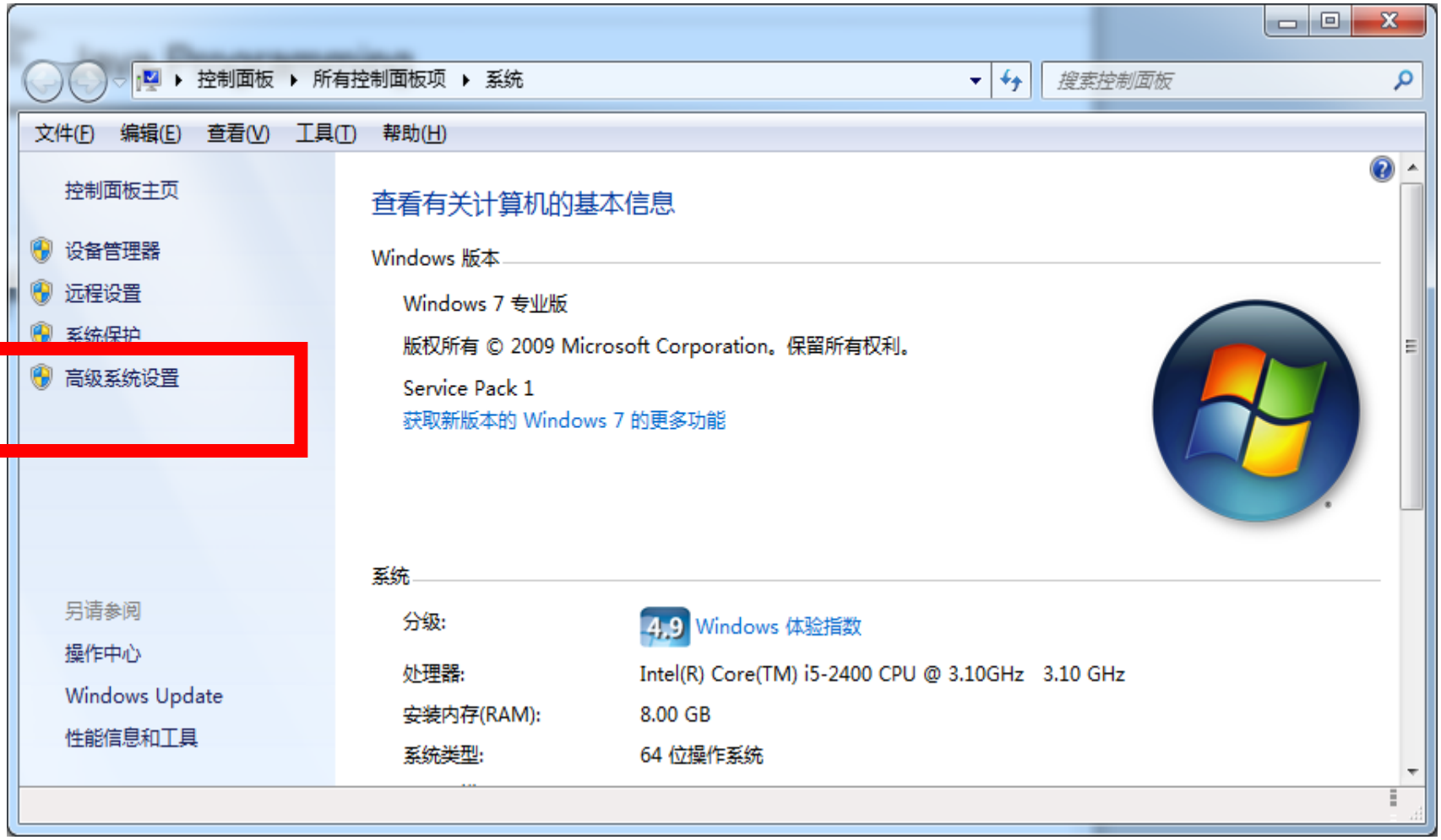
- 完成安装



JDK Deployment

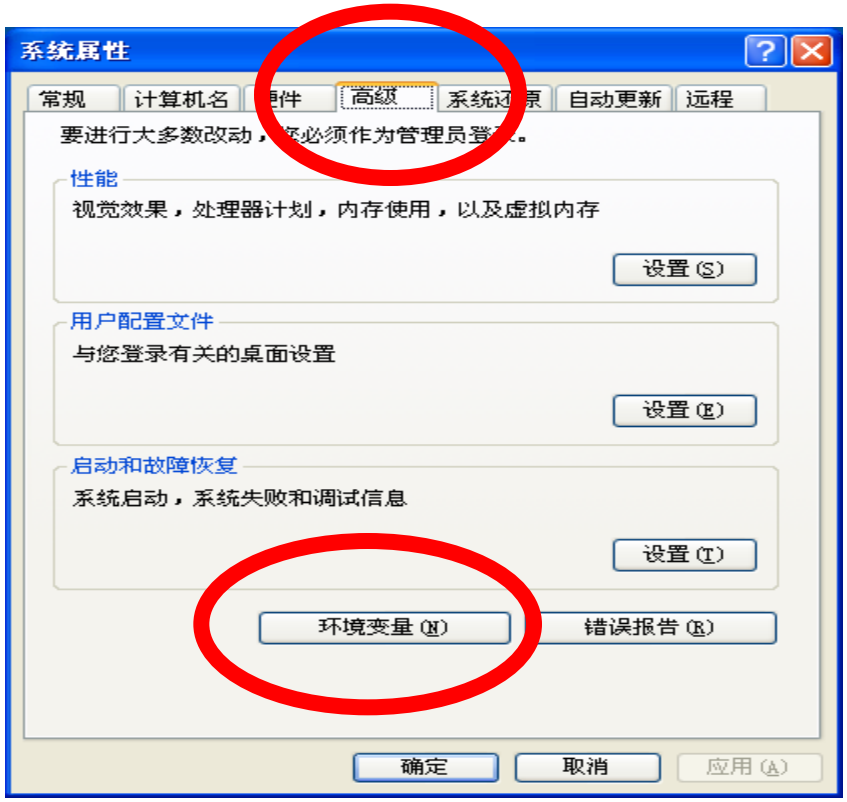
- 右键单击“我的电脑”，选择“属性”

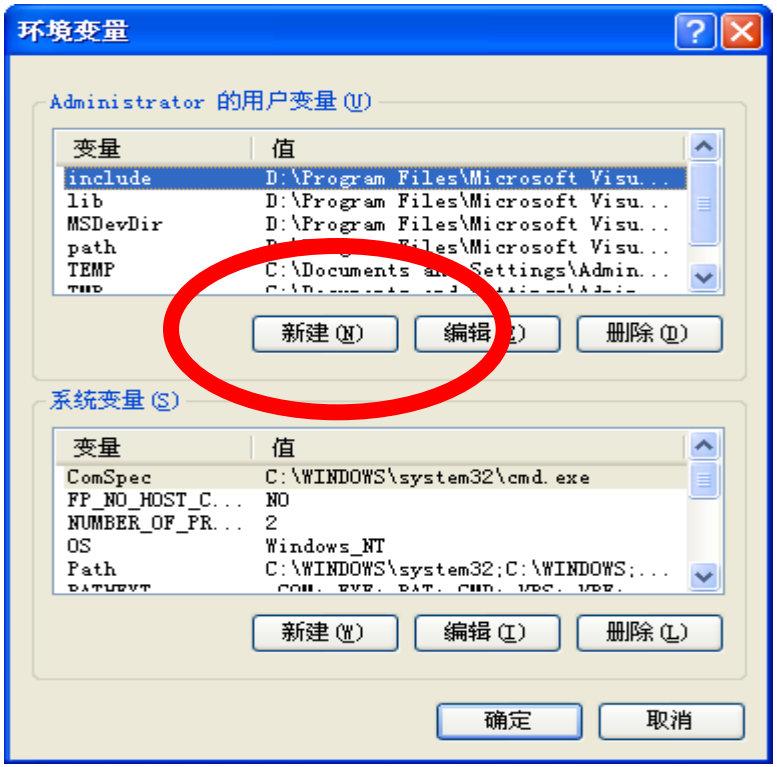



控制面板 ▸ 所有控制面板项 ▸ 系统
搜索控制面板
文件(F) 编辑(E) 查看(V) 工具(T) 帮助(H)
控制面板主页
设备管理器
远程设置
系统保护
高级系统设置
另请参阅
操作中心
Windows Update
性能信息和工具
查看有关计算机的基本信息
Windows 版本
Windows 7 专业版
版权所有 © 2009 Microsoft Corporation。保留所有权利。
Service Pack 1
获取新版本的 Windows 7 的更多功能
系统
分级:
4.9 Windows 体验指数
处理器:
Intel(R) Core(TM) i5-2400 CPU @ 3.10GHz 3.10 GHz
安装内存(RAM):
8.00 GB
系统类型:
64 位操作系统

JDK Deployment

- 选择“高级”选项卡，单击“环境变量”按钮

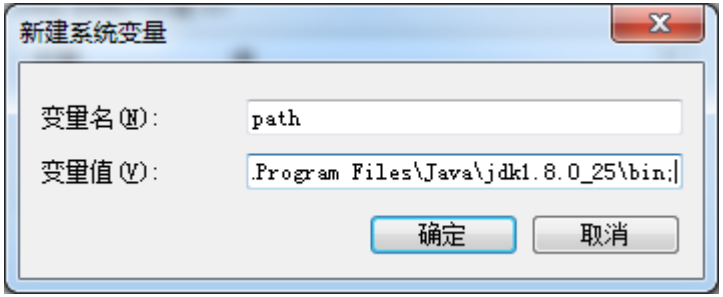




JDK Deployment

- 新建path变量

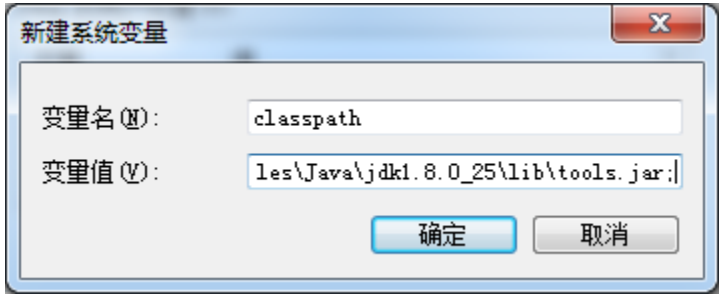
.; D:\Program Files\Java\jdk1.8.0_25\bin;



JDK Deployment

- 新建classpath变量

.; D:\Program Files\Java\jdk1.8.0_25\lib\tools.jar;



Environment Test

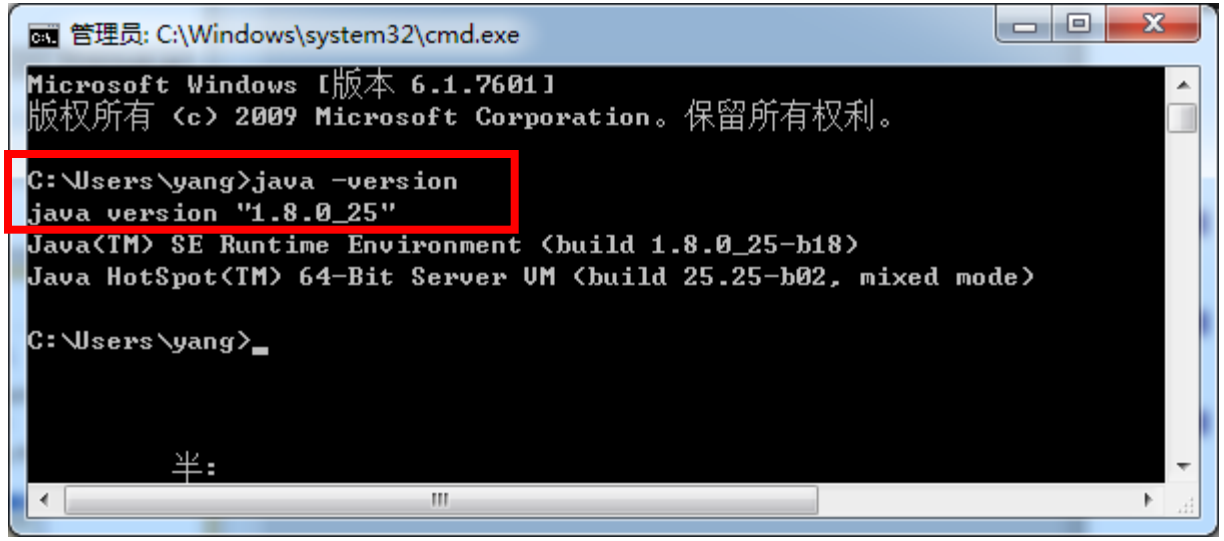
- 安装后，检测是否安装成功
- 【开始】→【运行】菜单
- 在运行框中输入cmd 并按下回车键



Environment Test

- 在打开的CMD窗口中输入命令

java -version



Your turn:

- After class repeat what I just did and some more.

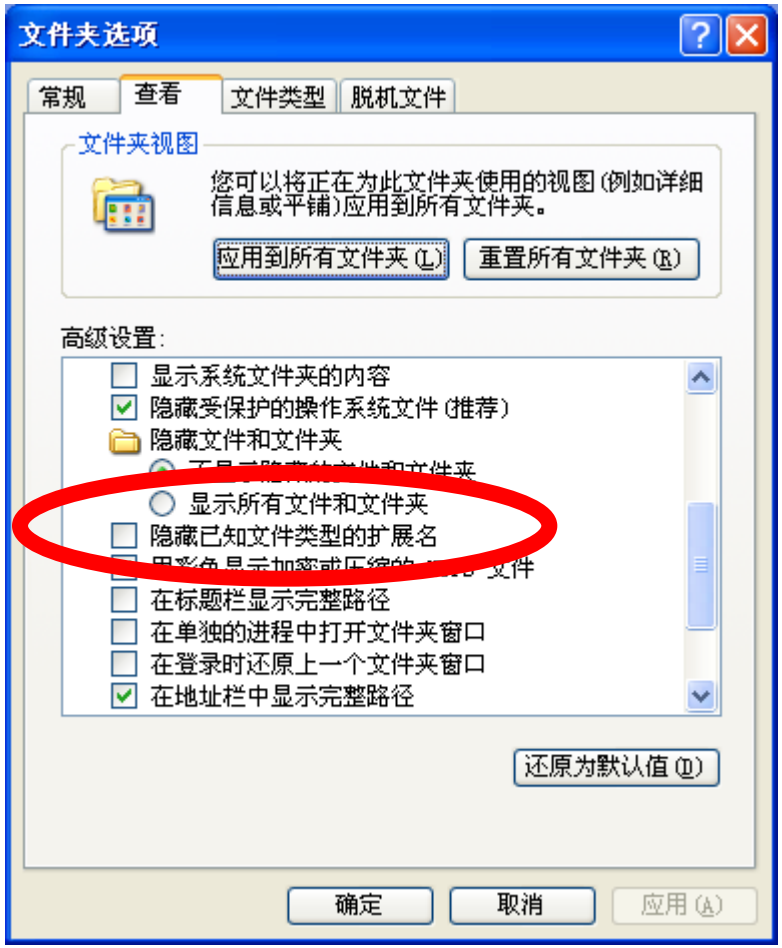
Prepared work



Prepared work

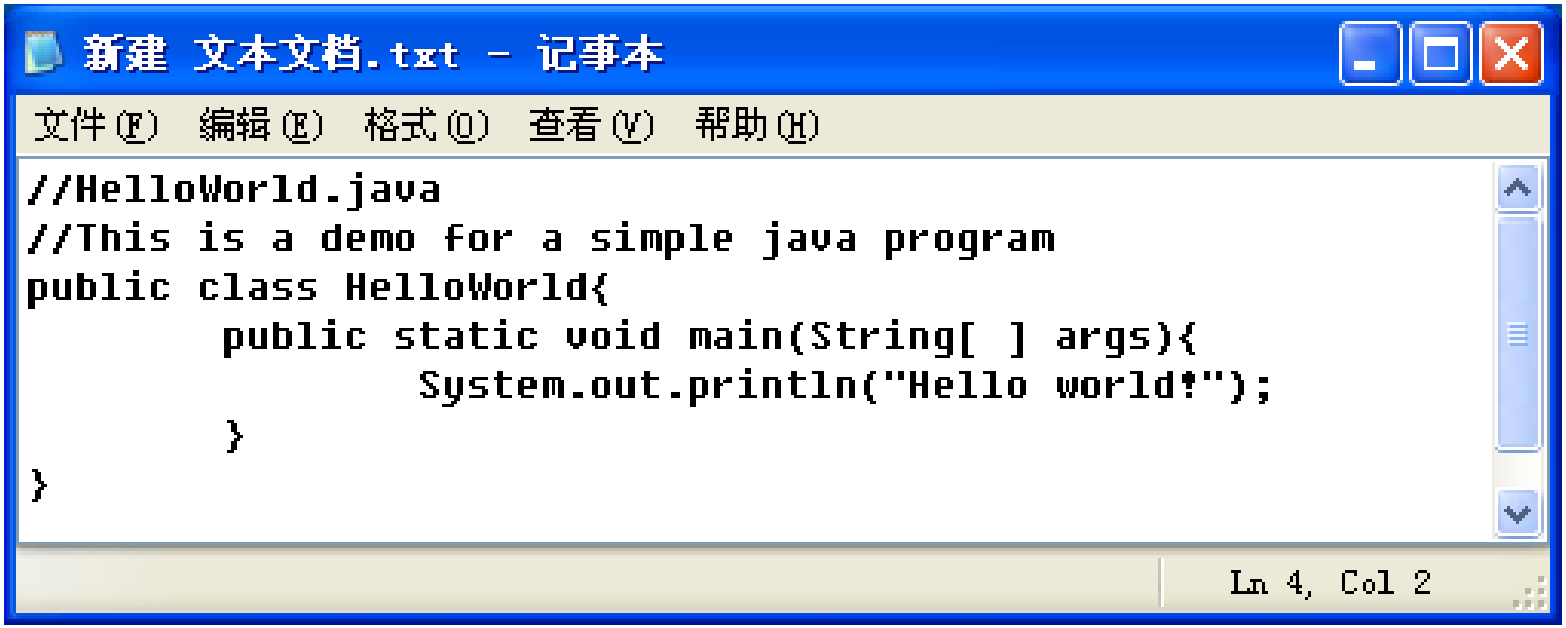


Prepared work



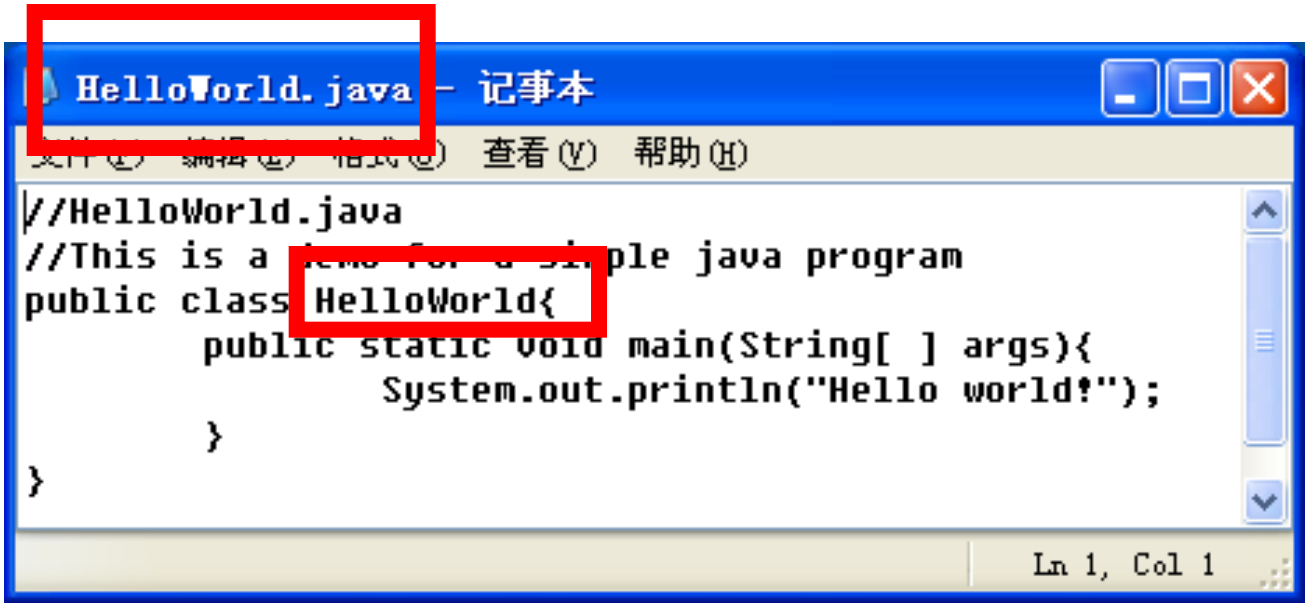
Your first Java program

- Create a new file using notepad



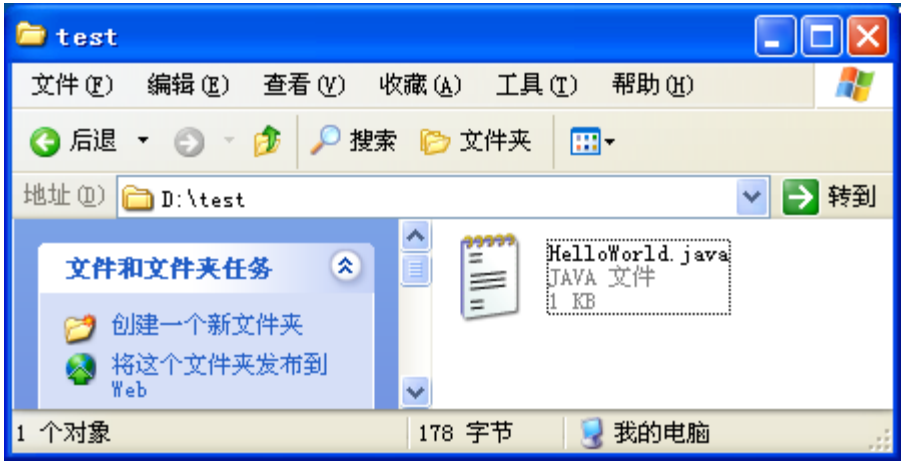
Your first Java program

- Choose “file"→"save as"
- Rename the file as **HelloWorld.java**

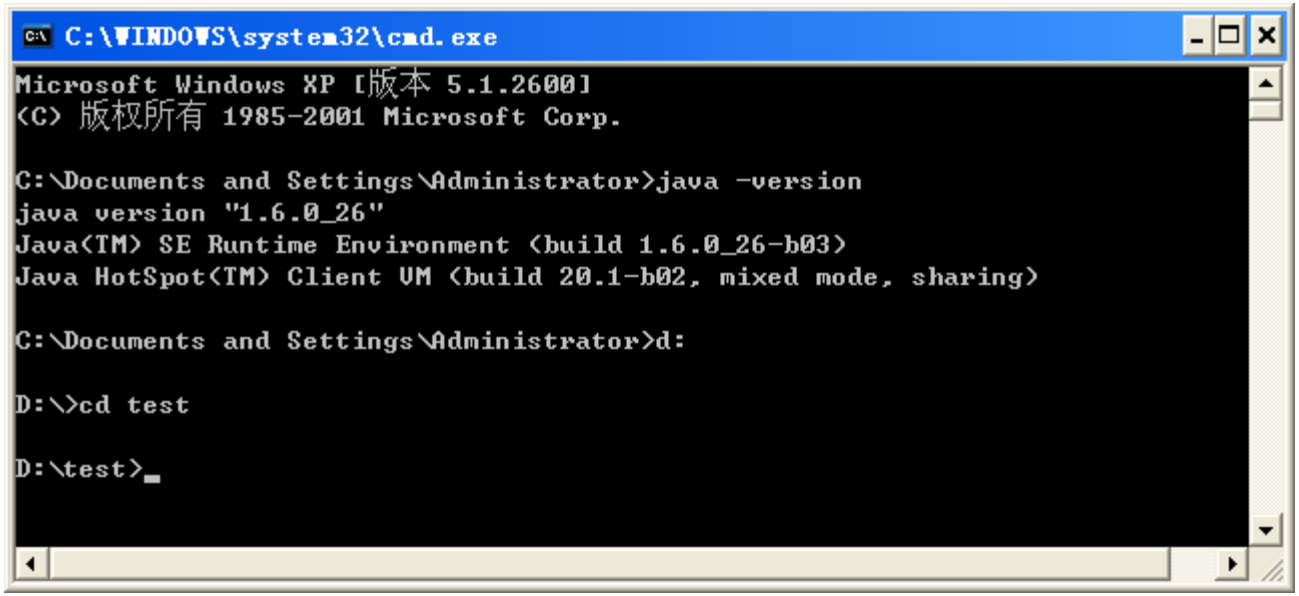


Your first Java program

- The file name must be the class name followed by the ".java" file-name extension.



Compile and run Java source file



Compile and run Java source file

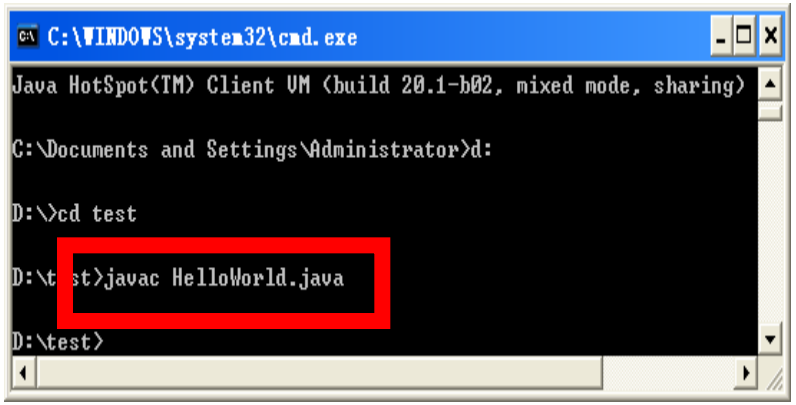
- Compile Java Source code

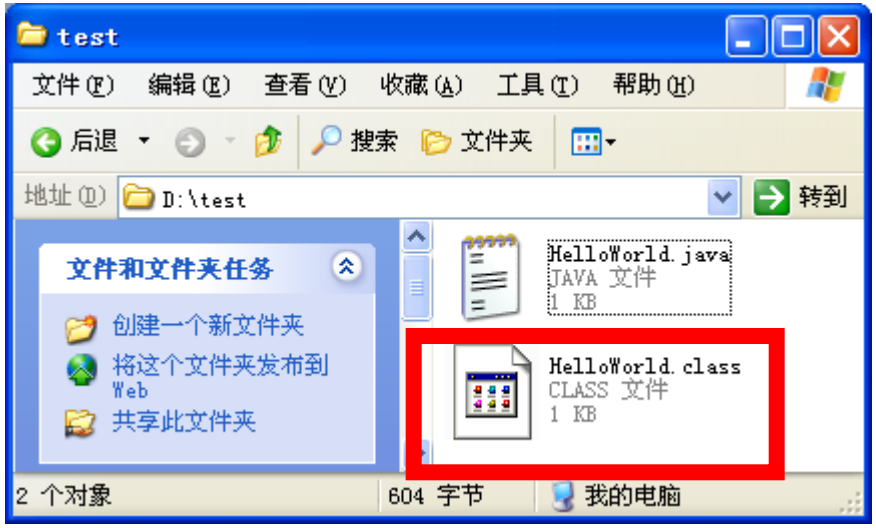
javac class_name.java

- 打开命令行窗口，进入程序所在目录
- 输入命令进行编译

Compile and run Java source file

- 输入javac命令，将源文件进行编译





Compile and run Java source file

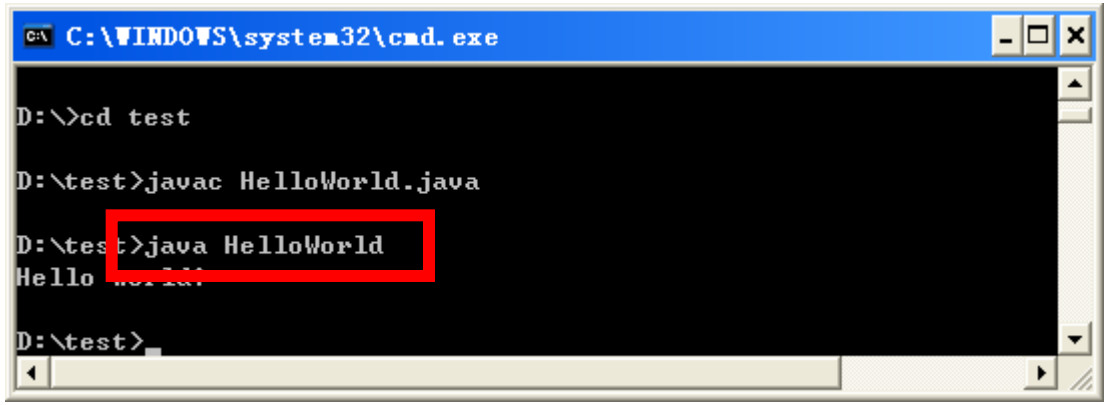
- Java解释程序
  - java解释javac所编译的字节代码

java 类文件名(不带扩展名) 参数

- 运行(run)
  - 输入命令运行程序
  - 观察程序运行结果

Compile and run Java source file

- 输入java命令，执行程序



Compile and run Java source file

- 操作很复杂
  - 编写源代码
  - 编译源代码 →javac命令
  - 运行程序→java命令

Analysis the HelloWorld Program

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

Declaring Java Technology Classes

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

Declaring Java Technology Classes

- Java is **object-oriented**
  - Everything is object.
  - Objects are the things in a program
  - Classes are BLUEPRINTS for designing these things
- Declaring Java Technology Classes
  - Java的程序最少必须包含一个您所声明的类

public class <class name>

Declaring Java Technology Classes

- a public Class Definition

```
public class name
```

  - 类名称必须与源文件名称一致

```
public class HelloWorld{
        ……
}
```

HelloWorld.java

Code block

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

The main Method

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

The main Method

- The main Method

```
public static void main(String[ ] args)
```

  - it's the entry point for your application and will subsequently invoke all the other methods required by your program

```
public static void main(String[ ] args){
}
```

Statements

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

Statements

- standard output

```
System.out.println();
```

```
System.out.print();
```

String

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
        System.out.println("Hello world!");
   }
}
```

String

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld{
   public static void main(String[ ] args){
      System.out.println("Hello"+"  world!");
       }
}
```

Source Code Comments

```
//HelloWorld.java
//This is a demo for a simple java program
public class HelloWorld {
   public static void main(String[ ] args){
       System.out.println("Hello world!");
   }
}
```

statement

- statement
  - 以分号(;)结束的一行或多行代码
  - 一条语句可以实现一个特定功能

```
System.out.println("Hello World!");
```

```
System.out.print ("Hello World!\n");
```

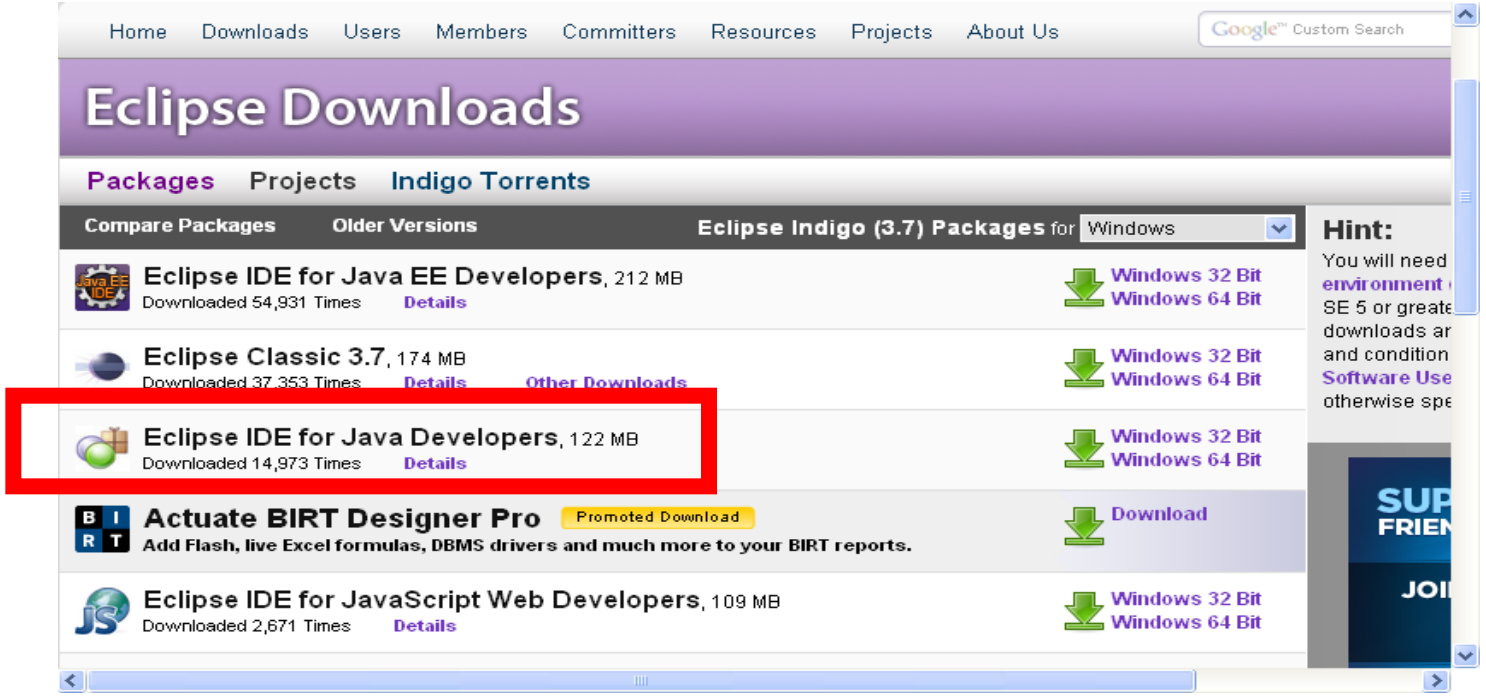
Eclipse World

- Eclipse
  - 为高度集成的工具开发提供一个全功能的、具有商业品质的工业平台
  - Eclipse项目
  - Eclispe工具开发
  - Eclispe技术项目

Download and install Eclipse

- Download Eclipse

http://www.eclipse.org/downloads/



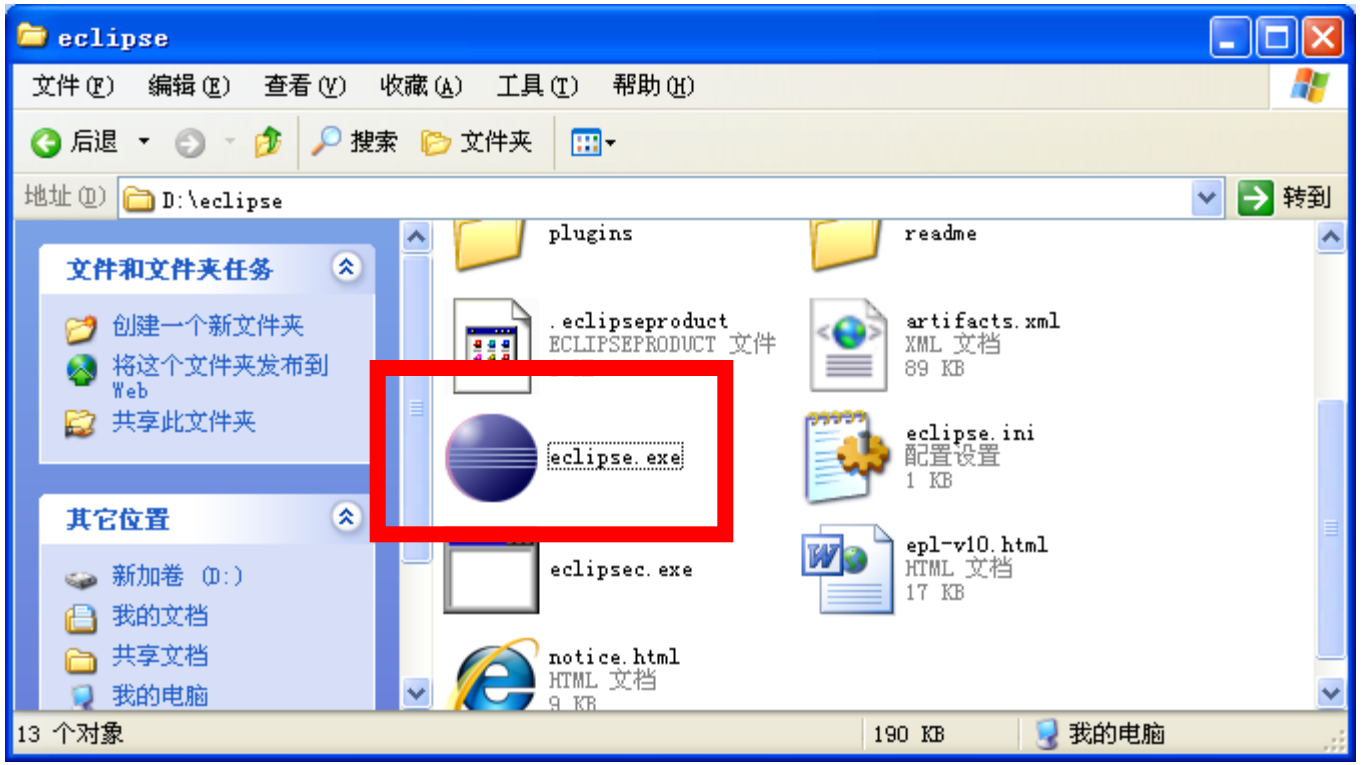
Download and install Eclipse

- Download Compression package

eclipse-java-indigo-win32.zip

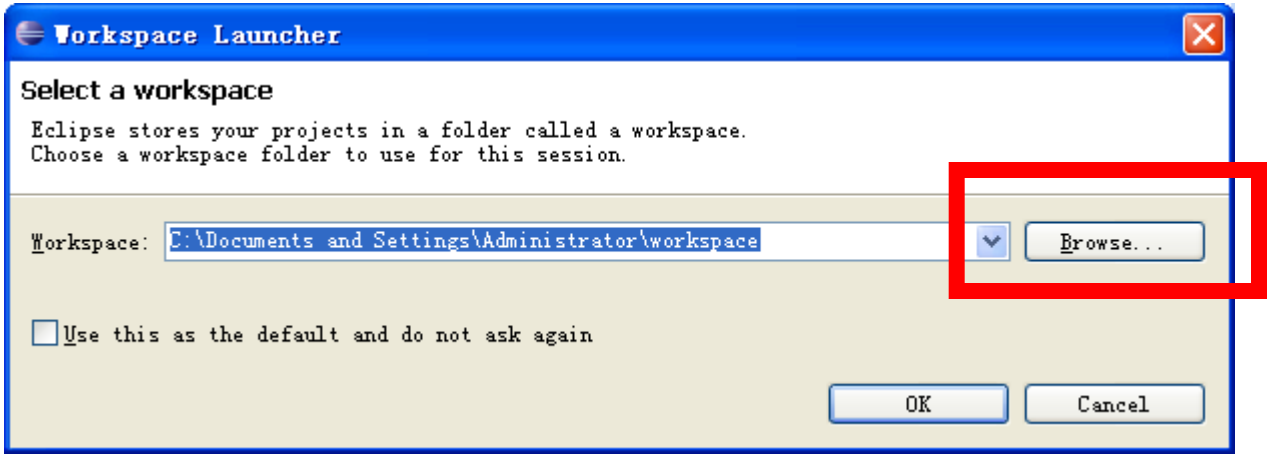
- Eclipse绿色的
  - 无须执行安装程序，解压压缩文件就可以用
  - 不过一定要先安装JDK

Run Eclipse


eclipse
文件(F) 编辑(E) 查看(V) 收藏(A) 工具(T) 帮助(H)
后退 搜索 文件夹
地址(D) D:\eclipse
转到
文件和文件夹任务
创建一个新文件夹
将这个文件夹发布到 Web
共享此文件夹
其它位置
新加卷 (D:)
我的文档
共享文档
我的电脑
plugins
readme
.eclipseproduct
ECLIPSEPRODUCT 文件
artifacts.xml
XML 文档
89 KB
eclipse.exe
eclipse.ini
配置设置
1 KB
eclipsec.exe
epl-v10.html
HTML 文档
17 KB
notice.html
HTML 文档
9 KB
13 个对象
190 KB
我的电脑

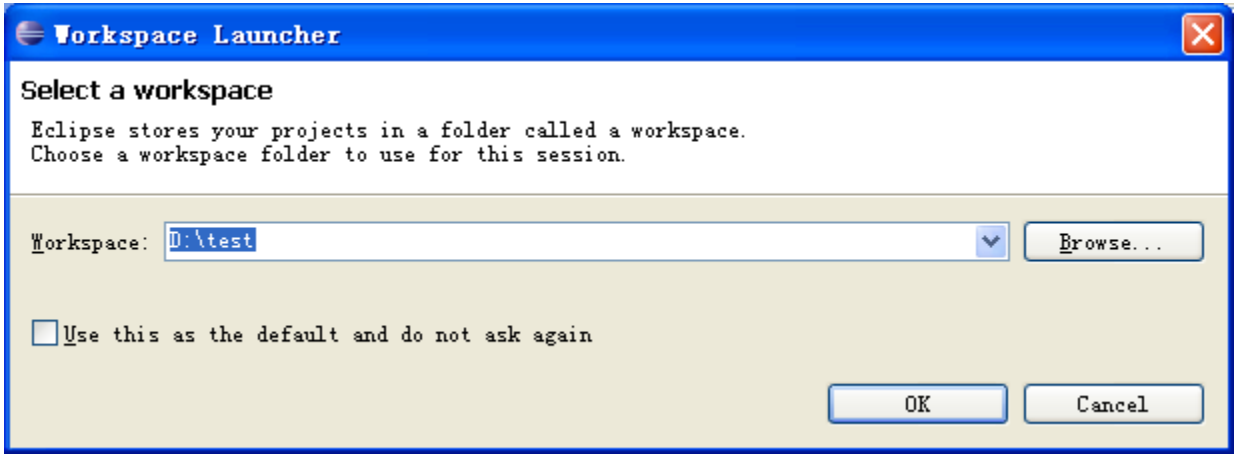
Run Eclipse


eclipse
INDIGO
(c) Copyright Eclipse contributors and others. 2000, 2011. All rights reserved. Eclipse is a trademark of the Eclipse Foundation, Inc. Oracle and Java are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

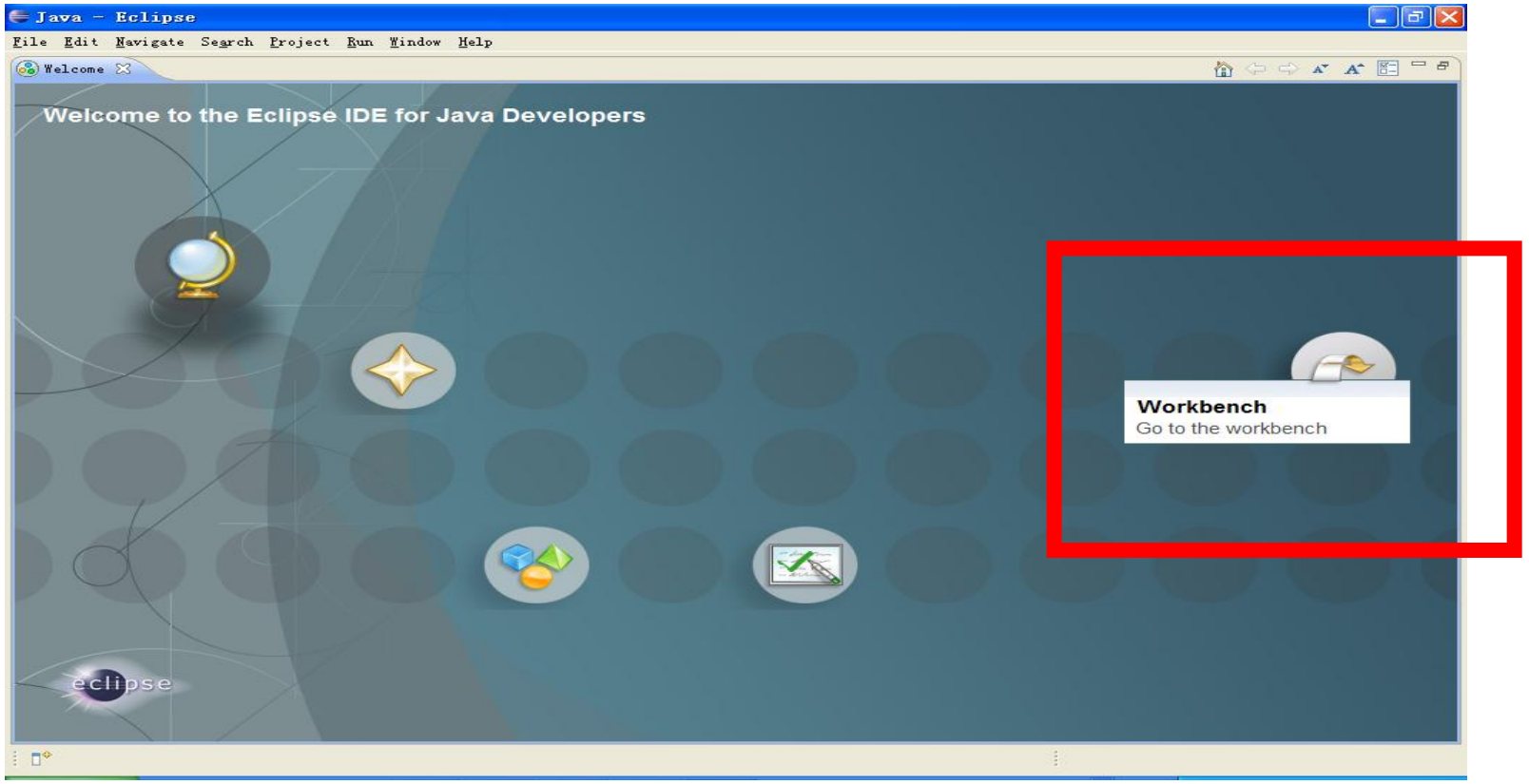

Workspace Launcher
Select a workspace
Eclipse stores your projects in a folder called a workspace.
Choose a workspace folder to use for this session.
Workspace:
C:\Documents and Settings\Administrator\workspace
Browse...
Use this as the default and do not ask again
OK
Cancel

Run Eclipse

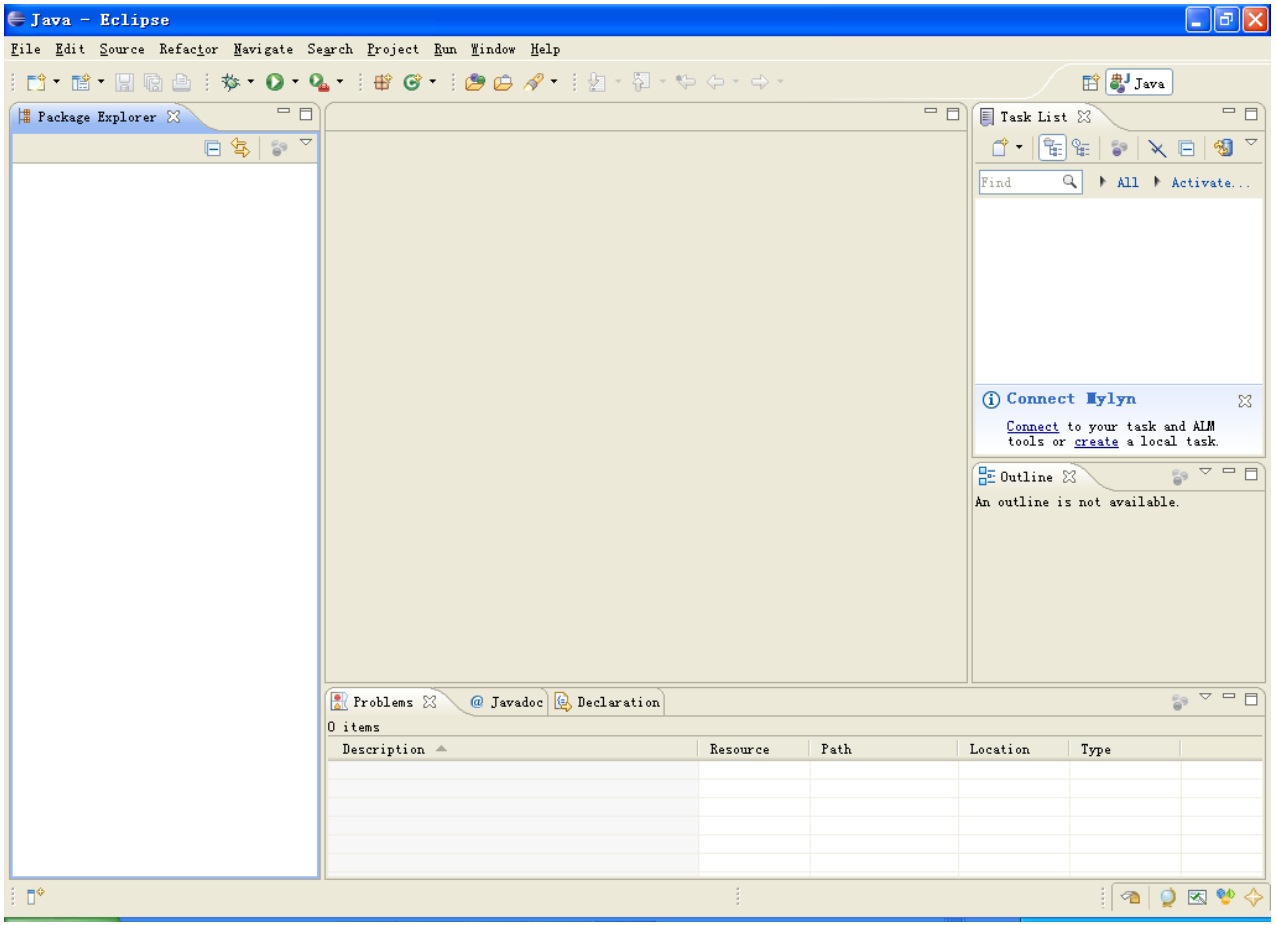
- Select a workspace



Run Eclipse

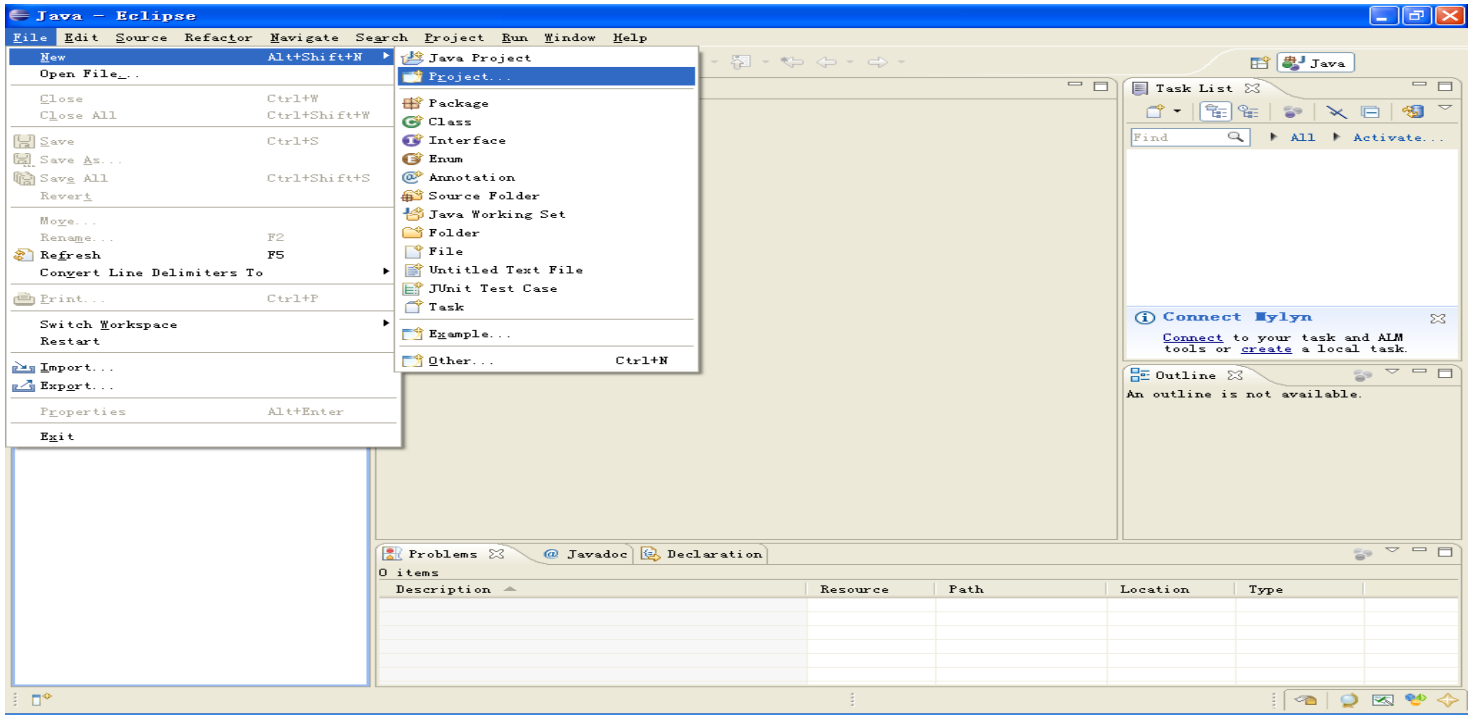

Java – Eclipse
File Edit Navigate Search Project Run Window Help
Welcome
Welcome to the Eclipse IDE for Java Developers
Workbench
Go to the workbench
eclipse

Run Eclipse

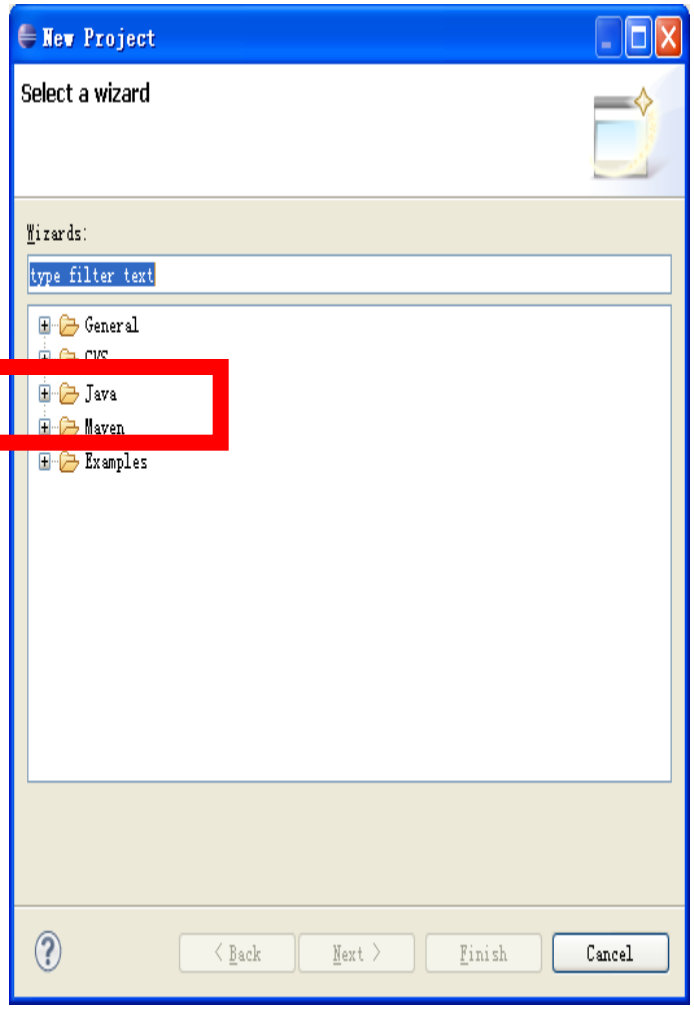

Java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Java
Package Explorer
Task List
Find
All
Activate...
Connect Mylyn
Connect to your task and ALM tools or create a local task.
Outline
An outline is not available.
Problems
Javadoc
Declaration
0 items
Description
Resource
Path
Location
Type

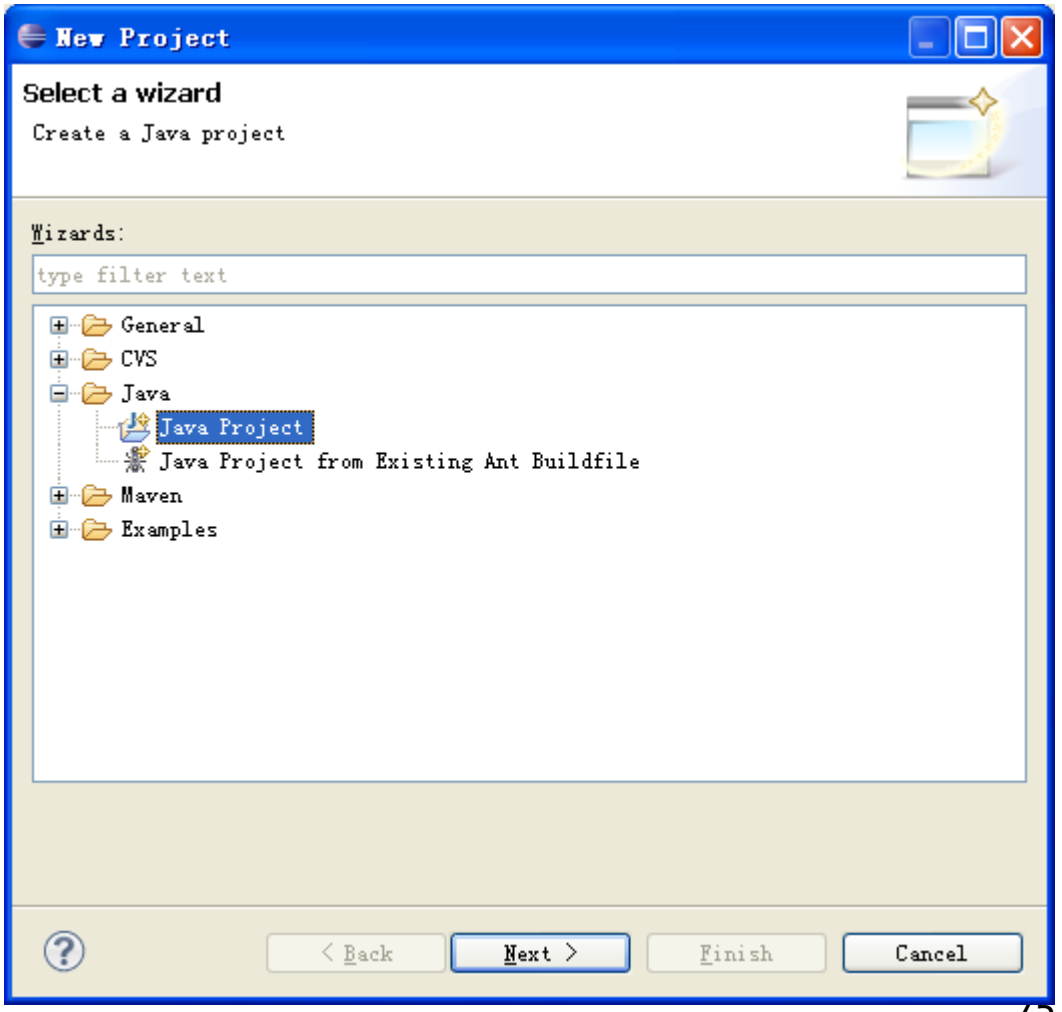
Run Eclipse

- Create a new Project
- 單击"File"→"New" →"Project"



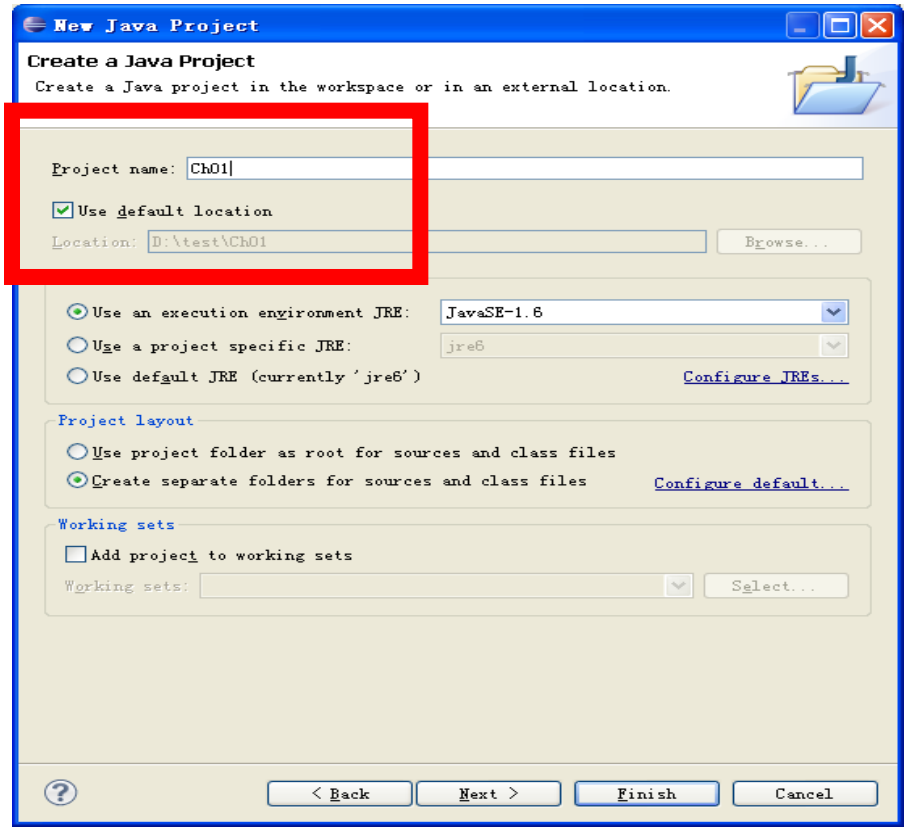
Run Eclipse


New Project
Select a wizard
Wizards:
type filter text
General
CVS
Java
Maven
Examples
Back
Next >
Finish
Cancel

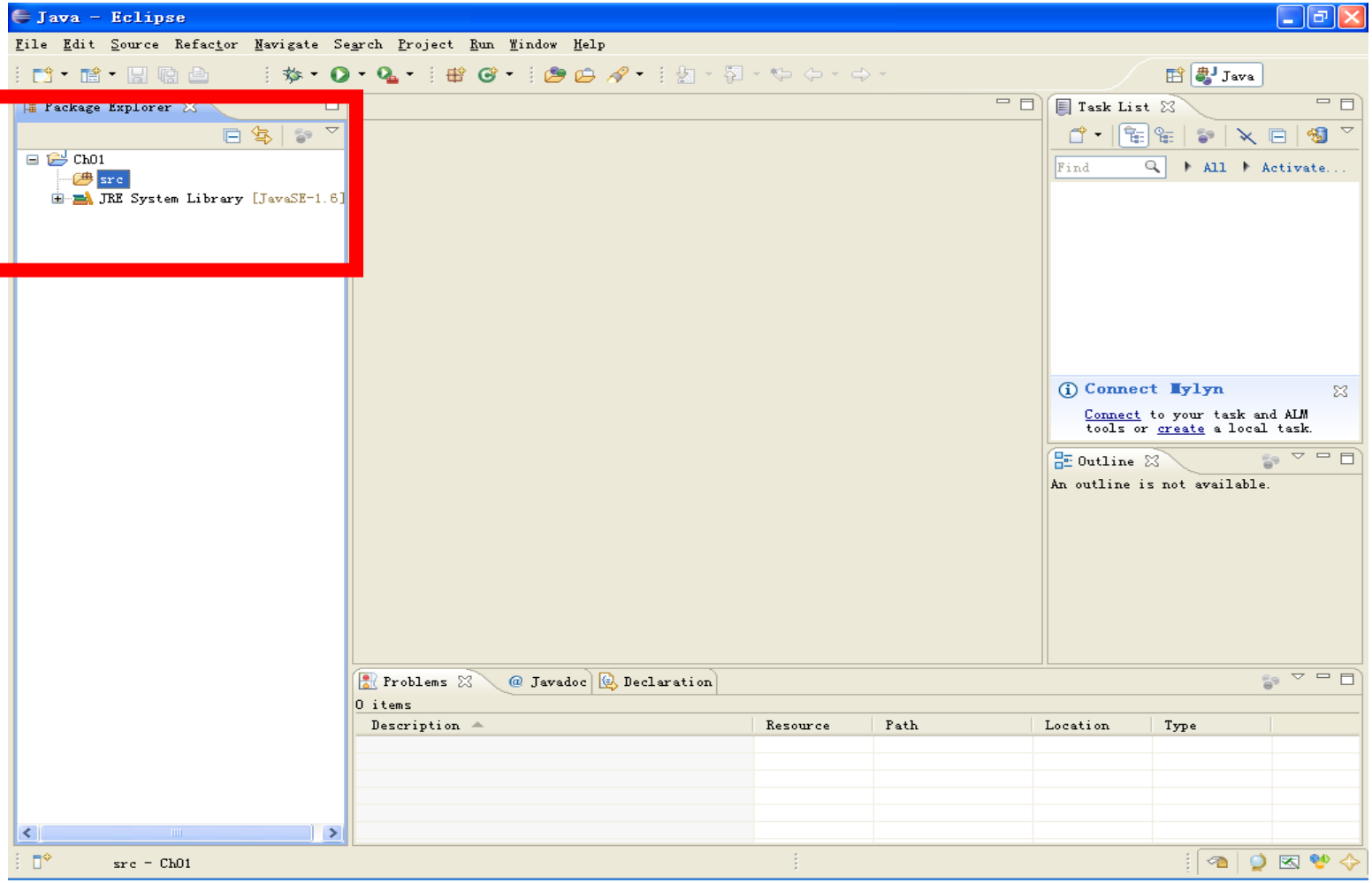

New Project
Select a wizard
Create a Java project
Wizards:
type filter text
General
CVS
Java
Java Project
Java Project from Existing Ant Buildfile
Maven
Examples
Back
Next >
Finish
Cancel

Run Eclipse

- 在“Project name”中输入项目名称，单击Finish



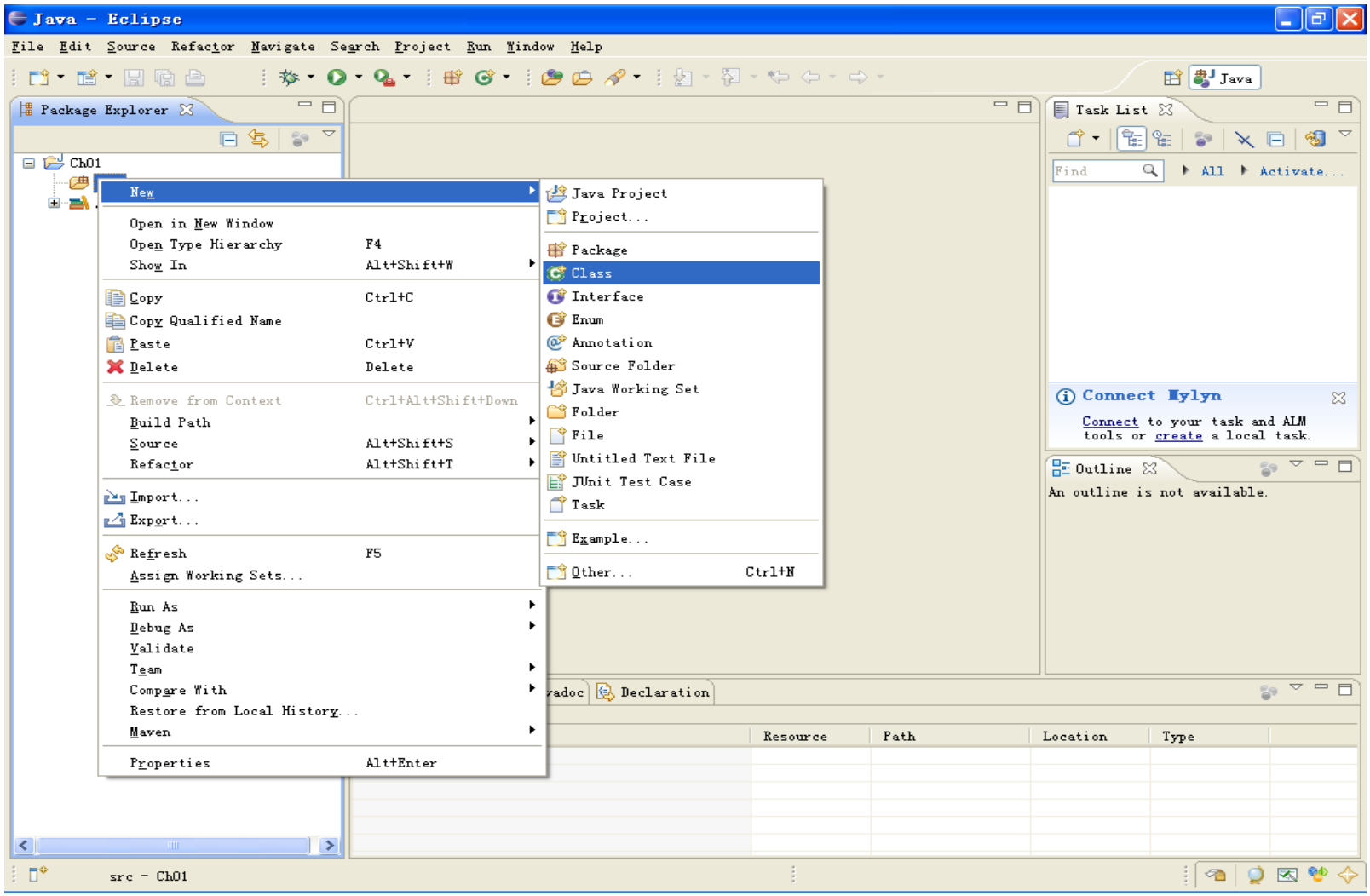
Run Eclipse


Java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Java
Package Explorer
Ch01
src
JRE System Library [JavaSE-1.6]
Task List
Find
All
Activate...
Connect Mylyn
Connect to your task and ALM tools or create a local task.
Outline
An outline is not available.
Problems
Javadoc
Declaration
0 items
Description
Resource
Path
Location
Type
src - Ch01

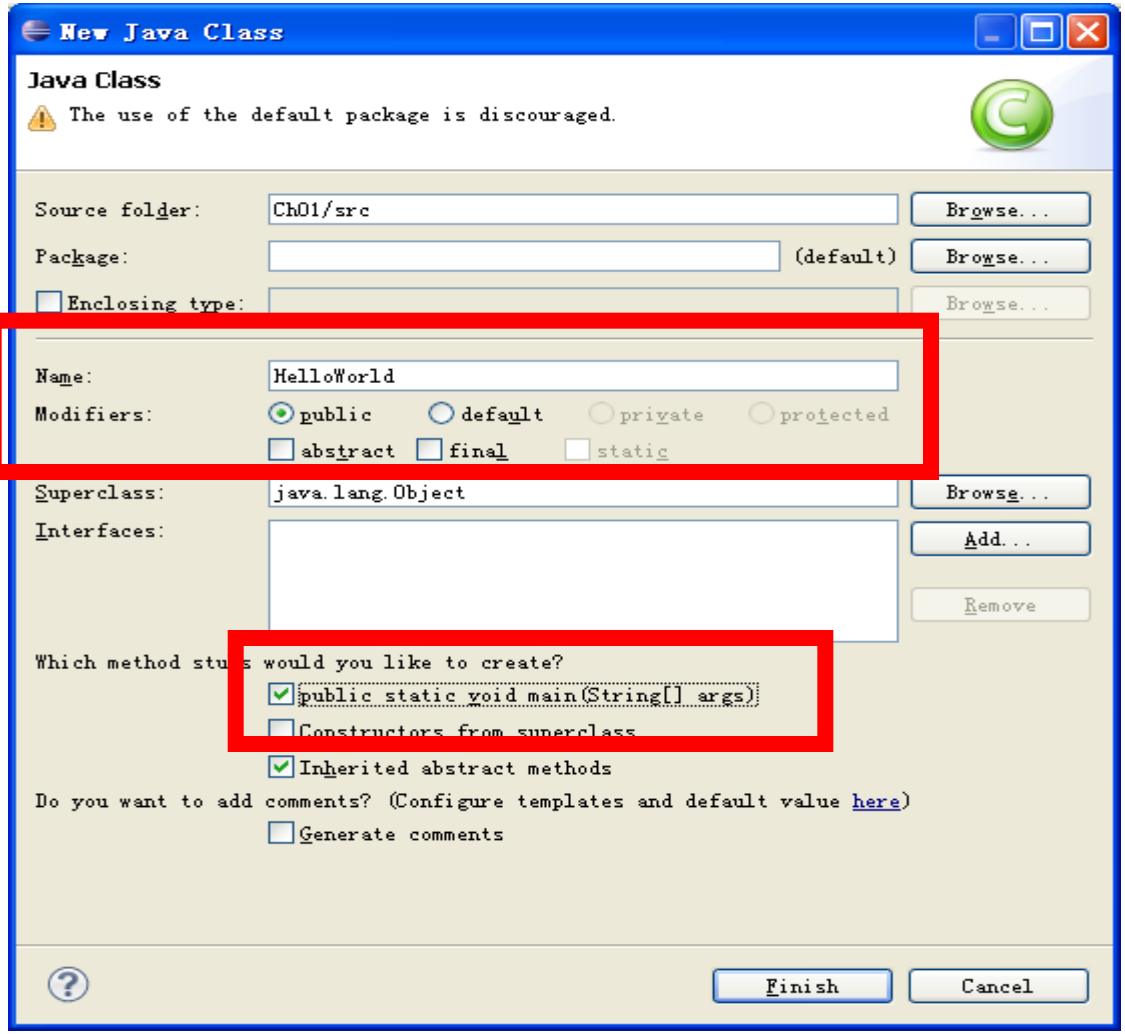
Run Eclipse

- 创建Java源文件
  - 类
- 在Eclipse的左侧双击”ch01”项目，选择“src”选项，单击鼠标右键，在弹出的快捷菜单中选择“New”→”Class”命令

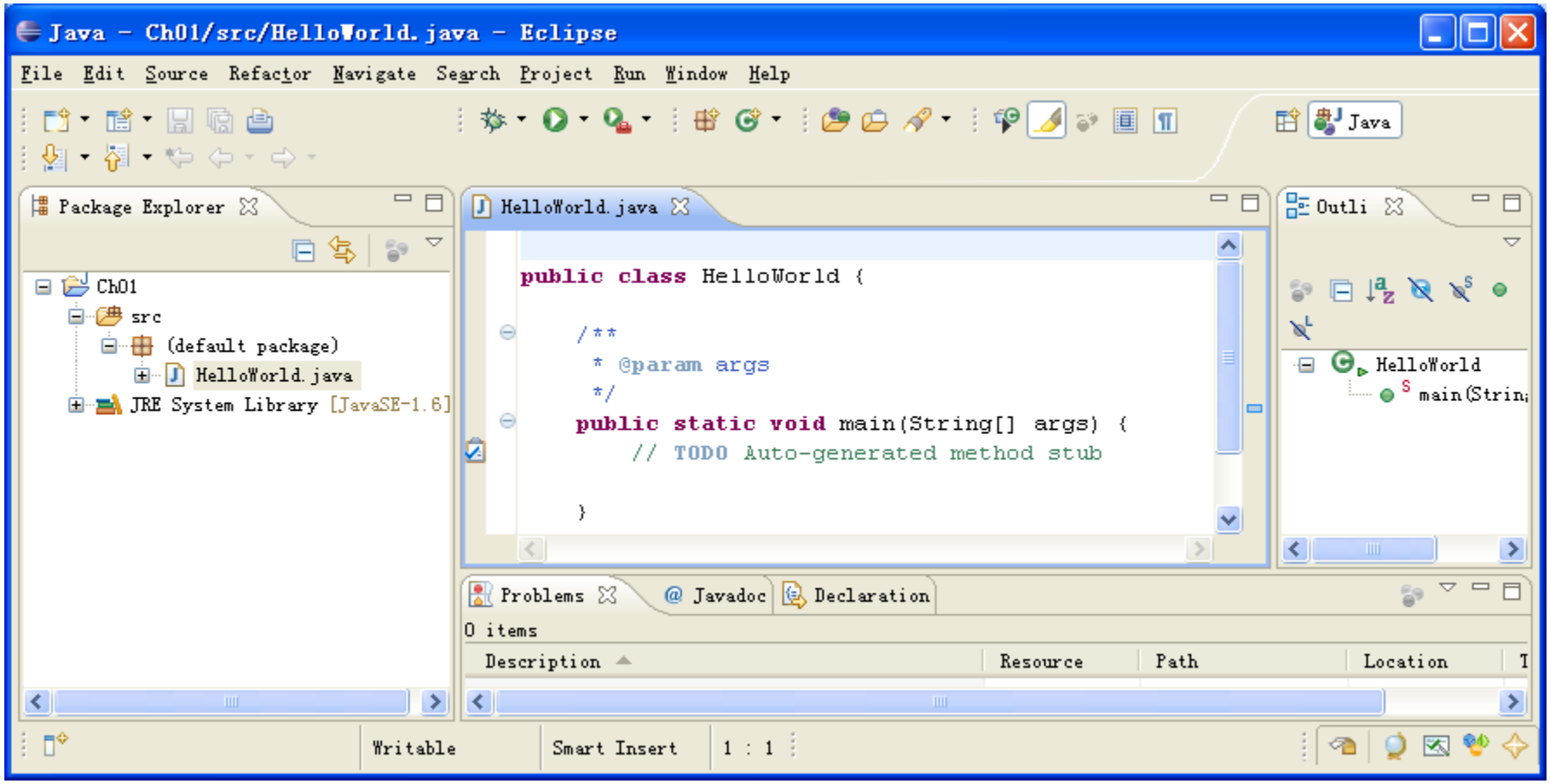
Run Eclipse


Java – Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
Ch01
New
Open in New Window
Open Type Hierarchy F4
Show In Alt+Shift+W
Copy Ctrl+C
Copy Qualified Name
Paste Ctrl+V
Delete Delete
Remove from Context Ctrl+Alt+Shift+Down
Build Path
Source Alt+Shift+S
Refactor Alt+Shift+T
Import...
Export...
Refresh F5
Assign Working Sets...
Run As
Debug As
Validate
Team
Compare With
Restore from Local History...
Maven
Properties Alt+Enter
Java Project
Project...
Package
Class
Interface
Enum
Annotation
Source Folder
Java Working Set
Folder
File
Untitled Text File
JUnit Test Case
Task
Example...
Other... Ctrl+N
Task List
Find
All
Activate...
Connect Mylyn
Connect to your task and ALM tools or create a local task.
Outline
An outline is not available.
Declaration
Resource
Path
Location
Type
src - Ch01

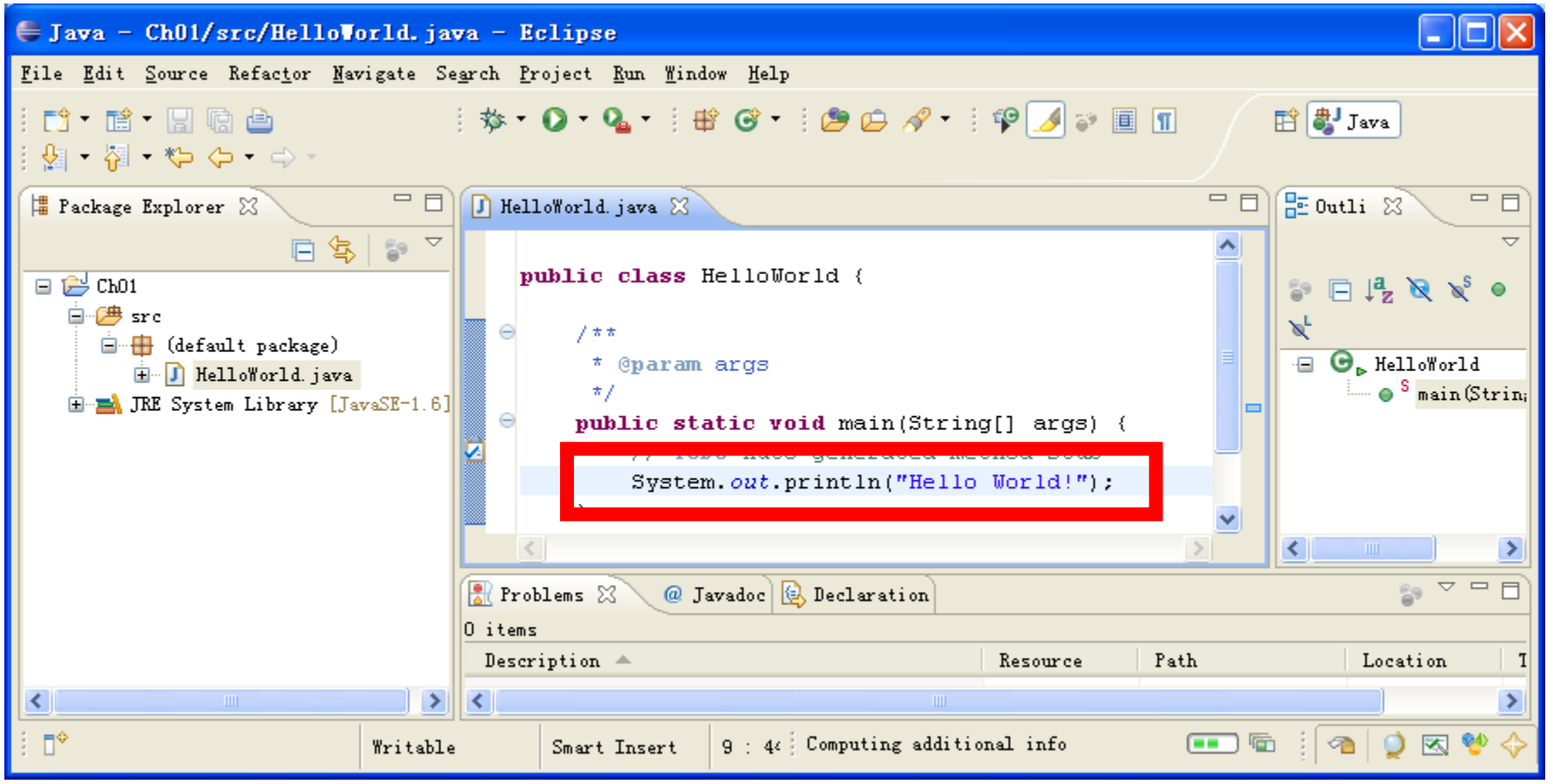
public class name



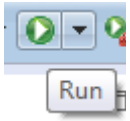
write HelloWorld program



write HelloWorld program

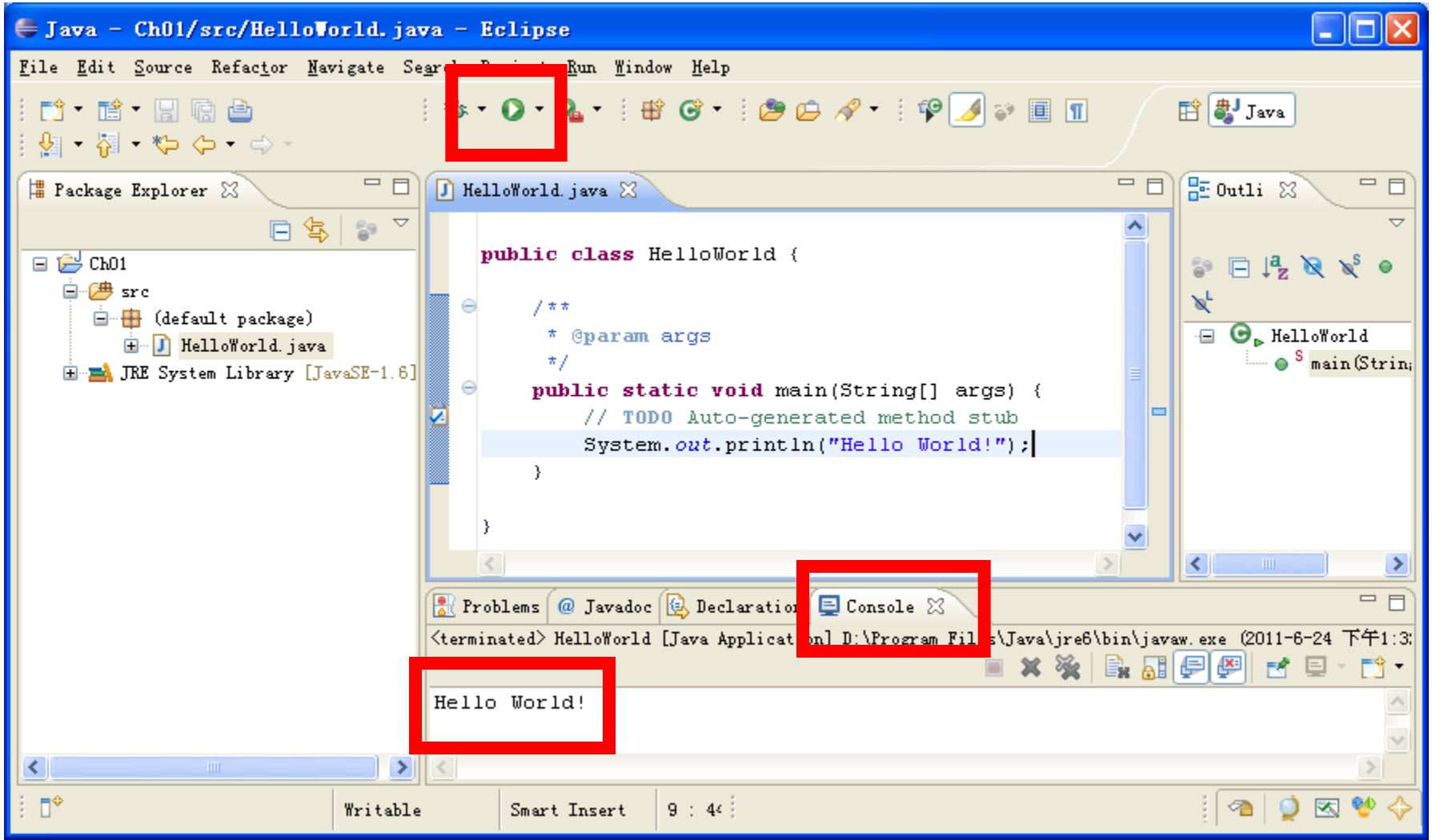


Compile and run Java source file

- Compile
  - Save the program
- Run
  - Press the run button 

  - ctrl+F11

Compile and run



The aims of this session

- Write and run your first programs!
- Use an integrated development environment (IDE) called Eclipse to help you do this…
  - I will talk a bit and then you have a go…
  - Demonstrators are here to help…
  - Work in pairs to help each other…